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Regime-Aware Stock Index Forecasting Under Latent Market States: A Hybrid Statistical Learning Framework with Cross-Market Validation

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Highlights

What are the main findings?

- A regime-aware statistical learning framework integrating Kalman filtering, Markov-switching dynamics, and nonlinear recurrent learning improves stock-index forecasting by combining noise reduction, latent-state identification, and nonlinear temporal adaptation.
- Regime-conditioned forecasting structures consistently outperform traditional and pure deep-learning models, with latent-state information providing the dominant contribution to predictive accuracy and economic value.

What are the implications of the main findings?

- Predictive accuracy alone does not ensure trading profitability; incorporating regime information is essential for translating forecasts into economic value.
- Model effectiveness varies across market environments, with regime-based forecasting capturing most predictability in developed markets, while hybrid statistical-learning structures provide additional gains in more volatile or emerging markets.

Abstract

This study proposes a hybrid forecasting framework that integrates Kalman Filtering (KF), Markov Switching (MS), and nonlinear recurrent learning for stock-index prediction. The KF component smooths short-term price noise, the MS model identifies latent return–volatility regimes, and the LSTM/GRU components learn nonlinear temporal patterns from regime-conditioned information. The framework is evaluated using the CSI 300, S&P 500, and Nikkei 225 indices through forecasting-accuracy measures, Bootstrap Diebold–Mariano tests with Modified Bayes Factor evidence, out-of-sample trading simulations, and robustness checks. The empirical results show that regime conditioning is the primary source of forecasting and economic improvement. KF–MS–LSTM performs best for the CSI 300 and Standard MS performs strongest for the S&P 500, while KF–MS–LSTM and KF–MS–GRU are more competitive for the Nikkei 225. In contrast, models without regime information, including pure LSTM/GRU and the standalone Transformer, generally exhibit weaker forecasting and trading performance. The findings suggest that latent market-state information is more important than neural-network complexity alone for robust financial forecasting, while the incremental value of Kalman filtering and recurrent learning remains market dependent. Overall, the results support regime-aware forecasting as an interpretable and economically meaningful approach for stock-index prediction under heterogeneous market environments.

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1. Introduction

Accurate stock price forecasting has long been a central challenge in financial research, driven by the complex, nonlinear, and regime-dependent dynamics of equity markets [1]. While the Efficient Market Hypothesis suggests that persistent return predictability should be limited in efficient markets [2], the Adaptive Market Hypothesis argues that market efficiency may vary over time as market environments, investor behavior, and institutional conditions evolve [3]. This perspective implies that financial return dynamics are inherently regime-dependent and motivate forecasting frameworks capable of adapting to changing latent market states rather than relying on a single stable return-generating process.

Traditional econometric models, such as the autoregressive integrated moving average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models, have been widely applied to financial forecasting but remain constrained by linear or weakly nonlinear structures that are often unable to capture structural breaks, volatility clustering, and nonstationary return dynamics [4,5]. These limitations become particularly important across heterogeneous financial markets, including the Chinese CSI 300, the United States S&P 500, and Japan's Nikkei 225, where differences in market maturity, information efficiency, and institutional environments generate distinct volatility regimes and information flows [6].

Recent advances in machine learning and deep learning have introduced flexible alternatives for modeling nonlinear temporal dependence on financial data [7]. Recurrent neural network (RNN) architectures, particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, have demonstrated strong performance in financial forecasting by preserving long-range sequential information through gating mechanisms [8,9]. LSTM-based models have been successfully applied to stock prediction tasks, including CSI 300 classification [10], gold price forecasting [11], and cross-index forecasting for the S&P 500 and NASDAQ [4], while GRU-based architectures often achieve comparable predictive accuracy with lower computational complexity [12]. Despite their flexibility, standalone deep-learning models remain vulnerable to overfitting under noisy or limited financial data and often produce unstable forecasts when trained directly on raw market series [13,14]. Recent machine-learning asset-pricing studies further show that increasingly complex architectures do not necessarily outperform simpler or regime-aware benchmarks, and that predictive gains depend strongly on market structure, data characteristics, and evaluation design [15,16].

Another important challenge in financial forecasting is market microstructure noise arising from bid–ask bounce effects, asynchronous trading, temporary liquidity shocks, and heterogeneous investor behavior [17,18]. Such short-term disturbances may obscure the latent return-generating process and reduce the reliability of downstream forecasting models. The Kalman Filter (KF) addresses this issue through recursive state estimation within a state-space framework, separating latent signals from transitory market noise [19]. In financial applications, Deepika and Nirupama Bhat [20] show that Kalman-filter preprocessing can improve LSTM-based forecasting performance by smoothing irregular market fluctuations before model estimation. However, the relationship between noise reduction and economic value is not necessarily monotonic; excessive filtering in highly volatile markets may suppress valuable information and generate unstable trading signals, particularly in the presence of transaction costs [21]. These considerations motivate the use of Kalman

filtering primarily as a preprocessing mechanism for regime identification rather than as a direct forecasting layer.

Stock markets are characterized by regime-dependent dynamics in which market conditions evolve across periods of high volatility, market stress, and relatively stable growth [22]. The Markov Switching (MS) framework captures such dynamics by allowing model parameters to vary endogenously across latent states without requiring pre-specified transition thresholds as in threshold autoregressive (TAR) or smooth transition autoregressive (STAR) models [19,23]. MS models have been widely applied to volatility regime detection [24], oil–stock dependence analysis [25], and cross-market risk–return heterogeneity [26]. However, most existing applications focus on single-market environments, leaving open the question of whether regime-aware forecasting structures generalize across heterogeneous financial systems without overfitting market-specific noise patterns.

To jointly exploit noise reduction, latent-state identification, and nonlinear temporal learning, recent studies have increasingly explored hybrid forecasting architectures. KF–LSTM combinations improve forecasting stability by filtering transitory market noise before recurrent learning [27,28], while MS–LSTM frameworks capture asymmetric return dynamics across bull and bear market phases more effectively than standalone deep-learning models [29,30]. Erlwein-Sayer et al. [31] further show that incorporating filtered regime probabilities as auxiliary LSTM inputs improves directional forecasting performance, whereas Sengupta et al. [32] document substantial forecasting gains from integrating Markov Switching structures with recurrent learning. In addition, the regime-switching literature increasingly emphasizes density and probabilistic forecasting for modeling financial uncertainty. Weighted likelihood-ratio approaches and regime-dependent density-forecast frameworks demonstrate that scenario-specific predictive distributions can improve uncertainty assessment under changing market conditions [33,34]. Although the present study focuses primarily on point forecasting and trading performance, probabilistic regime interpretation remains an important complementary component of the proposed framework.

Despite these advances, several important gaps remain in literature. First, many studies evaluate forecasting accuracy without clearly separating the incremental contributions of noise filtering, latent regime information, and nonlinear recurrent learning. Second, cross-market evidence remains limited, making it unclear whether hybrid forecasting frameworks generalize across heterogeneous institutional and volatility environments. Third, improvements in statistical forecasting accuracy do not necessarily translate into economic value, particularly in the presence of transaction costs and regime-dependent market risk [35–37].

To address these limitations, this study proposes a regime-aware hybrid forecasting framework integrating Kalman filtering, Markov Switching state identification, and recurrent neural learning. The proposed architecture operates in three stages. First, the Kalman Filter is applied to raw closing prices to generate noise-reduced filtered returns before regime estimation. Second, a Markov Switching model with Kim smoothing identifies three latent market states, namely adverse return–volatility, intermediate market, and favorable return–volatility regimes, and produces filtered regime probabilities together with a regime-weighted baseline signal. Notably, Kim-smoother is used only as the ex-post visualization tool; the out-of-sample forecasting experiment uses only forward-filtered regime probabilities based on current information. So, no future information enters the model estimation, hyperparameter selection, and forecasting. Third, recurrent neural networks receive raw returns, Kalman-filtered returns, regime probabilities, and the regime-weighted baseline component as forecasting inputs to model nonlinear temporal dependence and residual market adjustments. The residual-learning structure anchors forecasts around the

regime-weighted baseline while allowing the recurrent component to capture nonlinear deviations and short-run market interactions.

The empirical analysis examines three major equity indices: the CSI 300 (China), the S&P 500 (United States), and the Nikkei 225 (Japan). These markets differ substantially in market maturity, investor composition, institutional structure, and volatility dynamics, providing a suitable setting for cross-market validation. The proposed framework is compared against conventional econometric, volatility-based, pure deep-learning, regime-aware, and Transformer-based benchmark models. Forecasting performance is evaluated using RMSE, MAE, Theil's inequality coefficient (TIC), mean directional accuracy (MDA), R^2, Bootstrap Diebold–Mariano (DM) tests with Modified Bayes Factor (MBF) evidence, trading simulations, and robustness diagnostics.

This study pursues three main objectives. First, it provides a unified cross-market comparison of regime-aware and non-regime forecasting models across structurally heterogeneous equity markets. Second, it examines whether regime conditioning represents the primary source of forecasting improvement, while the incremental contributions of Kalman filtering and recurrent nonlinear learning remain market-dependent. Third, the study evaluates economic usefulness beyond statistical forecasting accuracy by incorporating trading performance, transaction-cost sensitivity analysis, and robustness diagnostics. Overall, the findings suggest that regime-state information, rather than neural-network complexity alone, plays the dominant role in robust stock-index forecasting.

The remainder of the paper is organized as follows. Section 2 presents the data, methodology, forecasting design, and evaluation procedures. Section 3 reports the empirical forecasting results, robustness analysis, and cross-market comparisons. Section 4 discusses the economic implications of the findings. Section 5 concludes the study.

2. Materials and Methods

This section presents the proposed regime-aware statistical learning framework for stock index forecasting under latent market states. The framework integrates Kalman filtering, Markov-switching state dynamics, and nonlinear recurrent learning within a chronologically consistent forecasting structure. The methodological design aims to capture observation noise, latent regime-dependent return dynamics, and nonlinear temporal dependence commonly observed in financial time series. The following subsections describe the data structure, stochastic modeling framework, forecasting architecture, robustness analysis, and evaluation procedures. Experiments were conducted in Python 3.12.7 using standard libraries, including TensorFlow and FilterPy, with minimal variation across runs, confirming robustness.

2.1. Data and Preprocessing

The empirical analysis employs daily closing prices from three major equity market indices representing heterogeneous market environments: the CSI 300 Index (China), the S&P 500 Index (United States), and the Nikkei 225 Index (Japan). The CSI 300 represents the Chinese A-share market, while the S&P 500 and Nikkei 225 provide developed-market benchmarks for cross-market validation. These markets are selected to evaluate the robustness and generalizability of regime-aware forecasting models across financial systems characterized by different levels of informational efficiency, volatility persistence, liquidity conditions, and regime-transition dynamics [2,15,38].

The sample period spans from January 2010 to December 2025. The data is obtained from the Yahoo Finance database. Nevertheless, the sample period, variable definitions, preprocessing procedures, and model implementation steps are fully documented to facilitate replication using comparable financial databases.

Let P_t denote the observed closing price at time t. Following standard practice in financial econometrics, continuously compounded returns are calculated as the natural logarithm of the ratio between consecutive closing prices. Financial return series commonly exhibit volatility clustering, nonlinear dependence, structural instability, and latent regime-switching behavior [19,39,40]. In addition, market prices may contain short-term noise arising from trading frictions and market microstructure effects [41,42]. These characteristics motivate a forecasting framework that integrates state-space filtering, latent-state inference, and nonlinear recurrent learning.

Before model estimation, all return series are standardized using statistics computed exclusively from the training sample. The standardized return is defined as $z_t = (r_t - \mu_{\text{train}})/\sigma_{\text{train}}$, where μ_{train} and σ_{train} denote the sample mean and standard deviation estimated from the training period. The same transformation parameters are subsequently applied to the validation and testing samples to preserve consistency across forecasting environments and prevent information leakage. Missing observations are handled through listwise deletion.

The full sample is partitioned chronologically into training, validation, and out-of-sample testing periods. Specifically, the first 80% of observations are used for model estimation and hyperparameter validation, while the remaining 20% are reserved for genuine out-of-sample forecast evaluation. To preserve chronological consistency, all model estimation, feature construction, and forecast generation procedures rely only on information available at the corresponding prediction date. Let

$$\mathcal{I}_t = \sigma(P_s, r_s, S_s : s \leq t) \tag{1}$$

denote the information set generated by historical market information up to time t, where S_t represents the latent market-state variable. This forecasting structure ensures consistency with real-time financial prediction environments and prevents look-ahead bias [43,44].

2.2. Kalman Filter for State Estimation

Financial asset prices may contain transitory noise arising from short-term trading frictions and temporary deviations from latent fundamental values, potentially obscuring the underlying return-generating process [41,42]. To mitigate these effects, this study employs a linear Gaussian Kalman Filter to extract a noise-reduced latent price signal before regime identification and nonlinear forecasting [45].

2.2.1. State-Space Specification

The Kalman Filter is formulated as a local-linear-trend state-space model. Let the latent state vector at time t be defined as

$$x_t = \begin{bmatrix} p_t^* \\ m_t \end{bmatrix}, \tag{2}$$

where p_t^* denotes the latent efficient price level and m_t the latent local trend component.

The latent-state evolution follows the transition equation

$$x_t = F x_{t-1} + w_t \quad , \quad w_t \sim \mathcal{N}(0, Q) \tag{3}$$

with the transition matrix

$$F = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}. \tag{4}$$

The observation equation linking the latent state to the observed market price is given by

$$P_t = Hx_t + v_t \quad , \qquad v_t \sim \mathcal{N}(0, R), \tag{5}$$

where $H = \begin{bmatrix} 1 & 0 \end{bmatrix}$. the observed disturbance v_t represents a noise measurement associated with observed closing prices. Q and R denote the process-noise and observation-noise covariance matrices, respectively. The initial latent state is specified as

$$x_0 = \begin{bmatrix} P_0 \\ 0 \end{bmatrix}, \tag{6}$$

where P_0 denotes the first observed training-sample price. The initial covariance matrix is defined as

$$P_{0|0} = \begin{bmatrix} R & 0 \\ 0 & 0.1R \end{bmatrix}. \tag{7}$$

To preserve the integrity of out-of-sample forecasting evaluation, all state-space parameters are estimated exclusively from the training sample. The observation-noise variance R is initialized using the variance of first differences in training-sample prices, while the process-noise covariance matrix is parameterized as

$$Q = \lambda R \tag{8}$$

where λ is selected from a prespecified candidate grid. The parameter configuration maximizing the Kalman-filter log-likelihood over the training sample is retained throughout the out-of-sample forecasting period.

2.2.2. Kalman-Filtered Returns

The Kalman Filter produces a filtered latent price estimate, denoted by $\hat{p}_t^*$. The corresponding Kalman-filtered log return, denoted by $\hat{r}_t^{KF}$, is defined as

$$\hat{r}_t^{KF} = \ln(\hat{p}_t^*) - \ln(\hat{p}_{t-1}^*), \tag{9}$$

where $\hat{p}_t^*$ and $\hat{p}_{t-1}^*$ denote the filtered latent price estimates at times t and $t-1$, respectively. The filtered return series provides a noise-reduced representation of the underlying market dynamics by attenuating short-term transitory fluctuations and measurement noise. Relative to raw returns, Kalman-filtered returns provide a smoother estimate of the latent return process and improve the stability of subsequent regime identification and nonlinear forecasting procedures.

2.3. Markov Switching Model for Regime Identification

Financial return dynamics often exhibit regime dependence characterized by changes in volatility, return persistence, and market conditions [19,40]. To capture these latent dynamics, this study estimates a Markov Switching (MS) model using the Kalman-filtered return series $\hat{r}_t^{KF}$.

Let $S_t \in \{1, 2, \ldots, K\}$ denote the latent market state at time t, where the regime process follows a first-order Markov chain. The regime-transition probabilities are defined as

$$\Pr(S_t = j \mid S_{t-1} = i) = p_{ij} \quad , \qquad \sum_{j=1}^{K} p_{ij} = 1, \tag{10}$$

where p_{ij} denotes the probability of transitioning from regime i at time $t-1$ to regime j at time t. The transition probabilities are collected in the transition matrix, as shown below:

$$P = \begin{bmatrix} p_{11} & \cdots & p_{1K} \\ \vdots & \ddots & \vdots \\ p_{K1} & \cdots & p_{KK} \end{bmatrix} \tag{11}$$

Conditional on the latent regime $S_t = k$, the Kalman-filtered return process is modeled as

$$\hat{r}_t^{KF} = \mu_k + \varepsilon_{t,k}\,, \;\; \varepsilon_{t,k} \sim \mathcal{N}(0, \sigma_k^2), \tag{12}$$

where μ_k and σ_k^2 denote the regime-specific conditional mean and variance, respectively. Under this specification, the conditional return distribution evolves according to the prevailing latent market state, allowing both the first and second moments of the return process to vary across regimes.

The number of latent regimes, K, is selected using training-sample information criteria. Models with $K = 2$, $K = 3$, and $K = 4$ are estimated, and the corresponding AIC and BIC statistics for the CSI 300 training sample are reported in Appendix A Table A1. The baseline specification adopts a three-regime structure representing adverse, intermediate, and favorable return–volatility states. The same $K = 3$ specification is imposed on the S&P 500 and Nikkei 225 indices to preserve cross-market comparability.

The Markov Switching model is estimated exclusively on the training-sample Kalman-filtered returns. During the out-of-sample forecasting period, the estimated transition matrix and regime-specific moments are held fixed, while regime probabilities are recursively updated using the Hamilton filter. The filtered regime probability is defined as

$$\pi_{t|t,k} = \Pr(S_t = k \mid \mathcal{I}_t), \tag{13}$$

where $\mathcal{I}_t$ denotes the information set available up to time t. These forward-filtered probabilities are subsequently incorporated into the forecasting model as regime-aware inputs.

For ex-post regime interpretation, Kim-smoothed probabilities are additionally computed as

$$\pi_{t|T,k} = \Pr(S_t = k \mid \mathcal{I}_T), \tag{14}$$

where $\mathcal{I}_T$ denotes the full-sample information set. Because smoothed probabilities condition on future observations, they are used exclusively for retrospective visualization and interpretation of the estimated regime structure. All forecasting experiments employ only forward-filtered probabilities from Equation (13), thereby preserving chronological consistency and avoiding look-ahead bias. All out-of-sample forecasts are generated exclusively using forward-filtered regime probabilities from the Hamilton filter. Kim-smoothed probabilities are used only for retrospective visualization and are never used in real-time forecast generation or trading simulation.

To evaluate the persistence, the expected duration of regime k is computed as

$$D_k = \frac{1}{1 - p_{kk}}, \tag{15}$$

where p_{kk} denotes the probability of remaining in regime k from one period to the next. Expected durations and transition probabilities provide a concise characterization of regime persistence and latent market-state stability.

2.4. Nonlinear Recurrent Statistical Learning

While the Markov Switching model captures latent regime-dependent dynamics, financial returns may exhibit additional nonlinear temporal dependence beyond regime transitions alone. To model these nonlinear patterns, the proposed framework incorporates recurrent statistical learning within a regime-aware forecasting structure.

Rather than treating recurrent neural networks as standalone prediction architectures, the proposed framework interprets recurrent learning as a nonlinear approximation of the conditional return-generating process under latent market states. The forecasting objective is to estimate the conditional expectation, as shown below:

$$\hat{r}_{t+1} = E(r_{t+1} \mid \mathcal{I}_t), \tag{16}$$

where $\mathcal{I}_t$ denotes the information set available up to time t. Under this formulation, the Markov-switching component captures low-frequency regime dynamics, while recurrent learning models nonlinear short-term temporal interactions. All forecasting procedures are implemented under a strictly chronological information structure to prevent look-ahead bias.

2.4.1. Regime-Aware Feature Construction

The recurrent forecasting model is constructed using a regime-aware feature vector that combines observed returns, Kalman-filtered returns, and filtered regime probabilities. Formally, let $\mathcal{F}_t = (r_t, \hat{r}_t^{KF}, \pi_t, \mu_t^{MS})$, where r_t denotes the observed return, $\hat{r}_t^{KF}$ is the Kalman-filtered return, π_t is the vector of forward-filtered regime probabilities, and μ_t^{MS} is the regime-weighted conditional mean defined as

$$\mu_t^{MS} = \sum_{k=1}^{K} \pi_{t|t,k} \mu_k. \tag{17}$$

This representation integrates short-term market fluctuations, latent price dynamics, and probabilistic market-state information into a unified forecasting input. To preserve chronological consistency, all regime probabilities used in $\mathcal{F}_t$ are forward-filtered probabilities computed using only information available up to time t.

2.4.2. Nonlinear Conditional Forecasting Model

The proposed forecasting framework combines regime-dependent market dynamics with nonlinear recurrent learning to estimate the conditional return-generating process. Let $\mathcal{F}_t$ denote the regime-aware feature vector defined in the previous subsection. The one-step-ahead forecast is modeled as

$$\hat{r}_{t+1} = \mu_t^{MS} + g_\theta(\mathcal{F}_t), \tag{18}$$

where μ_t^{MS} represents the regime-weighted conditional mean implied by the Markov Switching model, and $g_\theta(\mathcal{F}_t)$ denotes the nonlinear recurrent learning component parameterized by θ. Under this specification, the Markov-switching component provides a regime-dependent baseline forecast that captures low-frequency market-state dynamics, while the recurrent learning component models nonlinear short-term deviations and temporal interactions not explained by the latent-state process alone.

The predicted return is subsequently transformed into a one-step-ahead price forecast according to

$$\hat{P}_{t+1} = P_t \exp(\hat{r}_{t+1}), \tag{19}$$

where P_t denotes the observed price at time t. The forecasting structure from Equation (18) can alternatively be interpreted as a residual-learning specification. Specifically, the predicted nonlinear residual adjustment relative to the Markov-switching baseline is defined as

$$\hat{e}_{t+1} = \hat{r}_{t+1} - \mu_t^{MS} = g_\theta(F_t). \tag{20}$$

where $\hat{e}_{t+1}$ represents the nonlinear adjustment learned from the regime-aware sequential input structure. This decomposition shows that the proposed framework combines a regime-dependent statistical baseline with a nonlinear residual correction learned by the recurrent network. In this way, the forecasting framework integrates probabilistic regime dynamics with nonlinear temporal learning within a unified conditional forecasting system.

2.4.3. Recurrent Learning Architecture

To model nonlinear temporal dependence in financial returns, the proposed framework employs recurrent learning architectures based on Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks. Within the proposed regime-aware forecasting system, these architectures serve as nonlinear conditional approximation mechanisms operating on sequential regime-aware information.

Given the regime-aware input sequence $\mathcal{X}_t$, the recurrent learning component recursively updates a hidden-state representation according to

$$h_t = f_\theta(\mathcal{X}_t, h_{t-1}), \tag{21}$$

where h_t denotes the hidden-state vector and $f_\theta(\cdot)$ represents the recurrent transition function parameterized by θ.

For the baseline three-regime specification ($K = 3$), the input vector at time t is defined as

$$X_t = [r_t, \hat{r}_t^{KF}, \pi_{t|t,1}, \ldots, \pi_{t|t,K}, \tilde{r}_t^{MS}], \tag{22}$$

A rolling lookback window of length L is used to construct the sequential input representation,

$$\mathcal{X}_t = [X_{t-L+1}, X_{t-L+2}, \ldots, X_t]. \tag{23}$$

The baseline specification adopts $L = 5$, corresponding to one trading week, while alternative lookback windows are evaluated in the sensitivity analysis.

For the LSTM specification, the hidden states are updated through the gated recursions:

$$\begin{aligned}
i_t &= \sigma(W_i X_t + U_i h_{t-1} + b_i), \\
f_t &= \sigma(W_f X_t + U_f h_{t-1} + b_f), \\
o_t &= \sigma(W_o X_t + U_o h_{t-1} + b_o), \\
\tilde{c}_t &= \tanh(W_c X_t + U_c h_{t-1} + b_c), \\
c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \\
h_t &= o_t \odot \tanh(c_t),
\end{aligned} \tag{24}$$

where i_t, f_t, and o_t denote the input, forget, and output gates, respectively; $\tilde{c}_t$ is the candidate cell state; c_t denotes the memory-cell state; and h_t is the hidden-state output. The functions $\sigma(\cdot)$, and $\tanh(\cdot)$ represent the sigmoid and hyperbolic tangent activation functions, respectively, while $\odot$ denotes element-wise multiplication.

As a computationally more parsimonious alternative, the GRU architecture updates the hidden state according to

$$\begin{aligned} z_t &= \sigma(W_z X_t + U_z h_{t-1} + b_z), \\ r_t &= \sigma(W_r X_t + U_r h_{t-1} + b_r), \\ \widetilde{h}_t &= \tanh(W_h(r_t \odot h_{t-1}) + U_h X_t + b_h), \\ h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \widetilde{h}_t, \end{aligned} \tag{25}$$

where z_t and r_t denote the update and reset gates, respectively. Relative to the LSTM structure, the GRU reduces the number of trainable parameters while retaining the ability to model nonlinear temporal dependence and sequential interactions [46]. Overall, the recurrent learning architectures operate as adaptive nonlinear refinement mechanisms that complement the latent-state dynamics captured by the Markov Switching component.

2.4.4. Model Estimation and Forecast Generation

The recurrent learning models are estimated by minimizing the mean squared forecasting error between predicted and realized one-step-ahead returns. Let r_{t+1} denote the realized return and $\hat{r}_{t+1}$ denote the corresponding forecast. The learning objective is defined as

$$\theta^* = \text{argmin}_\theta \frac{1}{N} \sum_{t=1}^{N} (r_{t+1} - \hat{r}_{t+1})^2, \tag{26}$$

where θ represents the set of trainable network parameters.

Model estimation is conducted using the Adam optimization algorithm. To reduce overfitting and improve out-of-sample generalization, dropout regularization and early stopping based on validation loss are employed during training.

Hyperparameter selection is performed exclusively on the training and validation samples to preserve the integrity of the forecasting experiment. The tuning procedure considers alternative configurations for hidden-unit dimensions, dropout rates, learning rates, batch sizes, and lookback-window lengths. The final hyperparameter specification is selected according to validation-sample forecasting performance and subsequently held fixed throughout the out-of-sample evaluation period. Detailed hyperparameter settings are reported in Appendix A Table A2.

Forecast generation follows a strictly chronological recursive forecasting structure. At each forecasting origin t, the model uses only information available up to time t to generate the one-step-ahead forecast $\hat{r}_{t+1}$. During the out-of-sample period, regime probabilities are recursively updated using the Hamilton filter, while the recurrent learning architecture updates the nonlinear conditional forecast using the evolving regime-aware input sequence.

To improve reproducibility and comparability across forecasting architectures, all models are estimated using fixed random seeds and identical chronological evaluation protocols. The LSTM and GRU specifications are therefore compared under the same forecasting design and information structure.

2.5. Chronological Forecasting Design

To ensure genuine out-of-sample evaluation, the entire forecasting framework is implemented under a strictly chronological forecasting design. The sample is partitioned according to time order, with the first 80% of observations used for model estimation, hyperparameter selection, and validation, while the remaining 20% are reserved exclusively for out-of-sample testing.

All preprocessing procedures are conducted within the same chronological framework. Feature-standardization parameters are estimated exclusively from the training sample

and subsequently applied unchanged to the validation and testing periods. Missing observations are handled through listwise deletion.

The Kalman Filter, Markov Switching model, and recurrent learning architecture are all estimated using only training-sample information. During out-of-sample evaluation, model parameters are held fixed, while filtered states and regime probabilities are recursively updated as new observations become available. Consequently, each one-step-ahead forecast $\hat{r}_{t+1}$ is generated using only information available up to time t, thereby preserving consistency with real-time financial forecasting environments and preventing look-ahead bias throughout the empirical evaluation procedure. These preprocessing, filtering, estimation, and forecasting procedures are implemented identically across all benchmark and proposed models to ensure comparability in the empirical evaluation.

The validation subset is selected chronologically from the training period and is used for hyperparameter tuning and early stopping. Test-sample observations are not used in feature scaling, state filtering, regime estimation, or recurrent-learning optimization. This forecasting protocol preserves temporal ordering, prevents look-ahead bias, and maintains consistency with real-time financial forecasting environments.

2.6. Benchmark Models

To evaluate the contribution of each component within the proposed forecasting framework, this study compares the KF–MS–LSTM model against a set of benchmark and ablation specifications. The benchmark and ablation design is intended to isolate the predictive contributions of regime identification, Kalman-based filtering, and nonlinear recurrent learning within the forecasting framework.

The first benchmark is the ARIMA model, which provides a conventional linear time-series forecasting reference based on historical return dynamics. The second benchmark is the standard Markov Switching (MS) model, which generates forecasts using regime-dependent conditional means without recurrent nonlinear learning. This specification isolates the predictive contribution of latent regime information.

The recurrent-learning benchmarks include standalone LSTM and GRU models estimated using raw return sequences without Kalman-filtered returns or regime probabilities. These models evaluate whether recurrent learning alone can capture nonlinear temporal dependence on financial returns.

To assess the incremental contribution of regime conditioning, the MS–LSTM and MS–GRU specifications augment the recurrent-learning models with forward-filtered regime probabilities obtained from the Markov Switching component. The KF–LSTM specification incorporates Kalman-filtered returns together with recurrent learning, thereby isolating the contribution of state-space filtering in the absence of explicit regime conditioning. Finally, the proposed KF–MS–LSTM and KF–MS–GRU frameworks combine raw returns, Kalman-filtered returns, filtered regime probabilities, and the regime-weighted baseline return signal within a unified regime-aware forecasting structure.

This ablation design provides a systematic basis for identifying whether forecasting gains arise primarily from regime information, Kalman-based filtering, nonlinear recurrent learning, or their interaction effects. To ensure comparability across specifications, all competing models are estimated under the same chronological forecasting protocol, training-validation-testing split, evaluation metrics, and economic trading rules.

2.7. Sensitivity Analysis

Hyperparameter selection is conducted exclusively within the training and validation samples to preserve the out-of-sample validity of the forecasting experiment. For the Markov Switching component, the number of latent regimes is selected $K \in \{2, 3, 4\}$

using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). The baseline specification adopts a three-regime structure representing adverse, intermediate, and favorable return–volatility states while maintaining a parsimonious latent-state representation.

For the recurrent-learning models, the primary hyperparameters include the lookback-window length, number of hidden units, dropout rate, learning rate, batch size, and maximum number of training epochs. The baseline specification adopts a lookback window of $L = 5$, corresponding to one trading week, which balances the ability to capture short-term temporal dependence against excessive sequence length and model complexity.

To evaluate the robustness of the empirical findings, sensitivity analysis is conducted by varying the principal model hyperparameters across alternative specifications. The analysis considers alternative numbers of latent regimes, lookback-window lengths, hidden-unit configurations, dropout rates, and learning rates across the three equity markets. This procedure evaluates whether the forecasting results remain stable under reasonable changes in model specification.

All sensitivity analyses are conducted under the same chronological forecasting protocol and out-of-sample evaluation framework used in baseline specification. The corresponding results are reported in Appendices B and C and provide additional evidence regarding the stability and robustness of the proposed regime-aware forecasting framework.

2.8. Regime Reliability and Persistence Diagnostics

Because regime information plays a central role in the proposed forecasting framework, the reliability and persistence of the estimated latent states are examined in detail. Since the Markov Switching model produces probabilistic latent-state classifications rather than directly observable regime labels, regime reliability is evaluated using transition probabilities, expected regime durations, spell-length distributions, and regime-specific return–volatility characteristics.

For each market, the estimated transition probability matrix summarizes the probability of remaining in or switching between latent states. In particular, the diagonal element p_{kk} measures the probability that the process remains in regime k from one period to the next. Based on these probabilities, the expected duration of regime k is computed using Equation (15), which represents the average number of trading periods that the market is expected to remain within a given regime before transitioning to another state. Longer expected durations indicate more persistent latent market conditions, whereas shorter durations imply more frequent regime switching.

In addition to expected durations, the empirical distribution of regime spell lengths is examined. A regime spell is defined as a consecutive sequence of observations assigned to the same most-likely latent state. The spell-length distribution, therefore, provides supplementary evidence regarding whether the estimated regimes correspond to persistent adverse return–volatility states, intermediate market conditions, and favorable return–volatility states rather than short-lived transitory fluctuations.

The economic interpretability of the estimated regimes is further evaluated using regime-specific conditional returns and volatilities. Adverse regimes are expected to exhibit relatively weak returns and elevated volatility, intermediate regimes moderate return–volatility dynamics, and favorable regimes stronger returns and more stable market conditions. Together, these diagnostics provide an empirical basis for assessing whether the statistically identified latent states correspond to economically meaningful market environments.

2.9. Probabilistic Forecasting Interpretation

Although the primary empirical analysis focuses on point forecasts, the Markov Switching component additionally provides a probabilistic characterization of latent market conditions. Rather than assigning each observation to a single deterministic regime, the model generates a vector of conditional regime probabilities that summarizes the likelihood of the market occupying each latent state at time t.

Under the proposed framework, the one-step-ahead predictive return distribution can be expressed as a regime-weighted mixture distribution:

$$p(r_{t+1} \mid \mathcal{I}_t) = \sum_{k=1}^{K} \pi_{t+1|t,k} \phi(r_{t+1}; \mu_k, \sigma_k^2), \tag{27}$$

where $\pi_{t+1|t,k}$ denotes the predicted probability of regime k at time $t+1$, μ_k and σ_k^2 represent the regime-specific conditional mean and variance, respectively. Under this specification, the regime-switching component serves not only as a source of regime-aware forecasting inputs but also as a probabilistic mechanism for characterizing forecast uncertainty under evolving market conditions.

Within the proposed framework, the predictive regime probabilities are incorporated into the recurrent-learning input structure, allowing the nonlinear forecasting component to condition on both observed return dynamics and probabilistic latent market states. This probabilistic representation is particularly relevant in financial markets, where transitions between adverse, intermediate, and favorable return–volatility states are often gradual and uncertain rather than fully observable.

2.10. Computational Complexity

The proposed KF–MS–LSTM framework consists of three primary computational components: Kalman filtering, Markov Switching estimation, and recurrent nonlinear learning. Let T denote the number of time observations, d the dimension of the Kalman state vector, K the number of latent regimes, F the number of input features, H the number of hidden units, L the lookback-window length, and E the number of training epochs.

The Kalman Filter has approximate computational complexity,

$$\mathcal{O}(Td^3), \tag{28}$$

because each recursive filtering step requires matrix operations on the state covariance matrix. In the present application, the state vector contains only two latent components, namely the efficient price level and local trend, so the filtering stage remains computationally lightweight.

The Markov Switching filtering procedure has approximate complexity,

$$\mathcal{O}(TK^2), \tag{29}$$

since regime probabilities must be recursively updated across all possible state transitions. Because the baseline specification adopts only three latent regimes, the regime-filtering component remains computationally tractable for daily-frequency financial data.

The recurrent-learning component constitutes the most computationally intensive stage of the framework. The approximate LSTM training complexity is

$$\mathcal{O}(ETLH^2), \tag{30}$$

where L is the sequence length. Consequently, computational cost increases with longer sequential windows, larger hidden-state dimensions, and extended training iterations.

In terms of memory requirements, the Kalman Filter stores latent-state vectors and covariance matrices, while the Markov Switching component maintains transition probabilities and filtered regime distributions. The recurrent-learning models additionally require storage for sequential inputs, hidden states, and network parameters during training.

Overall, the proposed framework remains computationally feasible for daily-frequency forecasting applications because the latent-state dimension, number of regimes, and sequential lookback structure remain relatively parsimonious. However, computational costs may increase substantially in higher-frequency forecasting environments or large-scale cross-sectional asset applications requiring repeated recursive re-estimation.

2.11. Evaluation Metrics and Economic Performance Measures

Forecasting performance is evaluated using both statistical accuracy measures and economic trading performance criteria. Because predictive accuracy alone does not necessarily imply economic usefulness, the empirical analysis considers multiple complementary evaluation metrics commonly employed in financial forecasting studies.

2.11.1. Statistical Forecasting Accuracy

The primary statistical evaluation metrics are the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE),

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(\hat{P}_t - P_t\right)^2}, \tag{31}$$

$$MAE = \frac{1}{n}\sum_{t=1}^{n}\left|\hat{P}_t - P_t\right| \tag{32}$$

where $\hat{P}_t$ and P_t denote the predicted and observed prices, respectively. Additional forecasting diagnostics include the coefficient of determination (R^2),

$$R^2 = 1 - \frac{\sum_{t=1}^{n}\left(\hat{P}_t - P_t\right)^2}{\sum_{t=1}^{n}\left(P_t - \overline{P}\right)^2}, \tag{33}$$

and Theil's inequality coefficient (TIC),

$$TIC = \frac{\sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(\hat{P}_t - P_t\right)^2}}{\sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(\hat{P}_t\right)^2}\frac{1}{n} + \sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(P_t\right)^2}}, \tag{34}$$

where lower TIC values indicate greater forecasting accuracy.

Directional forecasting performance is further evaluated using Mean Directional Accuracy (MDA),

$$MDA = \frac{1}{n}\sum_{t=1}^{n} b_t, \qquad b_t = \begin{cases} 1, & (\hat{P}_t - P_{t-1})(P_t - P_{t-1}) \geq 0 \\ 0, & \text{otherwise} \end{cases}, \tag{35}$$

which measures the proportion of periods in which the model correctly predicts the direction of market movements.

To statistically compare predictive accuracy across competing forecasting models, the empirical analysis additionally employs the Diebold–Mariano (DM) test. The DM framework evaluates whether differences in out-of-sample forecasting performance are

statistically significant under alternative forecasting specifications. To improve robustness under non-Gaussian and serially dependent financial returns, bootstrap-based inference procedures are also considered in the empirical comparison stage.

2.11.2. Economic Performance Evaluation

Because statistical forecasting improvements may not translate directly into trading profitability, the forecasting models are additionally evaluated using a simple long-only trading strategy based on predicted price direction:

$$Position_t = \begin{cases} 1, & \hat{P}_t > P_{t-1} \\ 0, & \hat{P}_t \leq P_{t-1} \end{cases}, \tag{36}$$

where a long position is taken only when the predicted price exceeds the previous observed price. The corresponding trading return is computed as

$$R_t^{strategy} = Position_t \cdot r_t - TC_t, \tag{37}$$

where r_t denotes the realized log return and TC_t represents transaction costs associated with portfolio rebalancing.

Based on the resulting strategy-return series, the empirical analysis reports cumulative return, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, win rate, turnover, Calmar ratio, and information ratio.

To further evaluate the robustness of economic performance, transaction-cost sensitivity analysis is conducted under alternative trading-cost assumptions. In addition, tail-exclusion diagnostics are performed by removing the largest positive and negative trading-return observations at selected thresholds. This procedure assesses whether the reported risk-adjusted performance is broadly distributed across the forecasting horizon or driven primarily by a limited number of extreme trading periods.

2.12. Integrated Regime-Aware Hybrid Framework

The proposed KF–MS–LSTM framework integrates Kalman filtering, Markov Switching regime identification, and recurrent nonlinear learning within a unified forecasting architecture. The Kalman Filter first extracts noise-reduced latent price signals and generates filtered returns from observed market prices. These filtered returns are then used in the Markov Switching component to estimate latent market regimes and forward-filtered regime probabilities representing adverse, intermediate, and favorable return–volatility states. Finally, the recurrent learning component combines raw returns, Kalman-filtered returns, regime probabilities, and the regime-weighted baseline signal to generate one-step-ahead return forecasts.

Figure 1 summarizes the hierarchical information flow of the proposed forecasting framework. The Kalman Filter serves as a state-estimation tool and noise-reduction mechanism, the Markov Switching model captures regime-dependent market dynamics, and the recurrent learning architecture models nonlinear temporal dependence beyond the latent-state process. Together, these components form a regime-aware nonlinear forecasting system designed for real-time financial prediction.

The empirical implementation follows a strictly chronological forecasting structure. All preprocessing procedures, parameter estimation, hyperparameter selection, and validation steps are confined to the training period, while out-of-sample forecasts are generated recursively using only information available at the prediction date. This design preserves temporal consistency and prevents look-ahead bias throughout the forecasting experiment.

In addition to forecast evaluation, the framework incorporates benchmark comparisons, sensitivity analysis, regime-persistence diagnostics, probabilistic forecasting interpretation, computational-complexity assessment, and economic trading evaluation. Collectively, these components provide a comprehensive basis for assessing the statistical accuracy, robustness, interpretability, and practical feasibility of the proposed regime-aware forecasting framework.

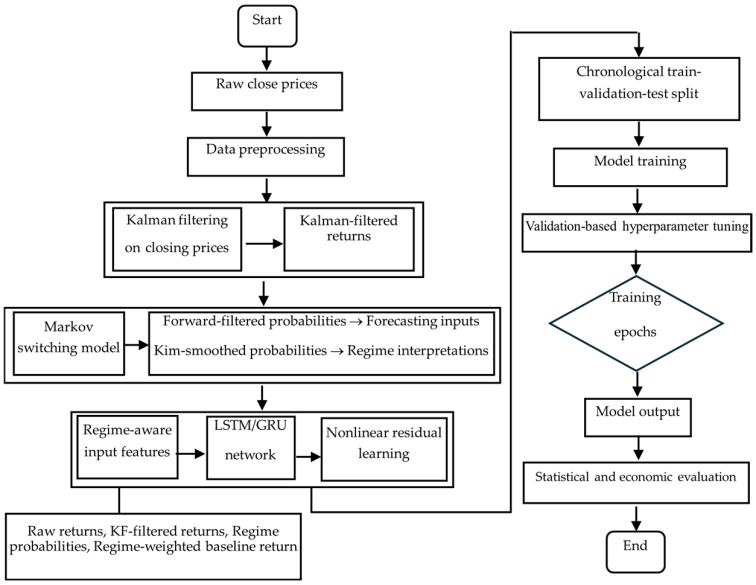


Figure 1. Integrated KF–MS–LSTM Regime-Aware Forecasting Framework.

3. Results

This section evaluates the statistical forecasting performance, robustness, and economic value of the proposed KF–MS–LSTM framework across different market environments. The empirical analysis examines whether integrating Kalman filtering, Markov Switching regime identification, and nonlinear recurrent learning improves stock-price forecasting relative to conventional econometric, volatility-based, recurrent-learning, and attention-based benchmarks.

The analysis proceeds in four stages. First, one-step-ahead forecasting accuracy is compared across competing benchmark models. Second, the study evaluates whether improvements in statistical forecasting performance translate into economically meaningful trading outcomes after accounting for transaction costs. Third, component-level and ablation analyses are conducted to isolate the incremental contributions of regime information, Kalman-filtered signals, and nonlinear recurrent learning. Finally, robustness analysis examines the stability of the results under alternative lookback windows, transaction-cost assumptions, tail-event exclusions, and cross-market validation settings.

The benchmark specifications include ARIMA, GARCH, standard Markov Switching (MS), Pure LSTM, GARCH–LSTM, MS–LSTM, KF–MS–LSTM, Pure GRU, MS–GRU, KF–MS–GRU, Transformer, and KF–MS–Transformer. This benchmark structure allows the empirical analysis to distinguish among the predictive contributions of volatility modeling, regime conditioning, Kalman-based filtering, recurrent nonlinear learning, and attention-based architecture.

All models are evaluated under the same chronological forecasting protocol. Statistical forecasting performance is assessed using RMSE, MAE, TIC, MDA, and R^2, while forecast-comparison robustness is examined using bootstrap Diebold–Mariano tests with Modified Bayes Factor (MBF) evidence classification. Economic evaluation is based on trading-

performance measures, including cumulative return, annualized return, Sharpe ratio, maximum drawdown, turnover, and transaction-cost sensitivity analysis.

3.1. Preliminary Data Characteristics

Before estimating the forecasting models, this subsection examines the statistical properties of the CSI 300 Index used in the empirical analysis. The analysis is based on daily closing prices and continuously compounded log returns over the sample period from January 2010 to December 2025. Descriptive statistics, the Augmented Dickey–Fuller (ADF) test, and the Ljung–Box test [47] are reported to evaluate distributional characteristics, stationarity, and serial dependence.

Table 1 reports the descriptive statistics for the CSI 300 Index closing prices and log returns. The average closing price is 3533.5693 with a standard deviation of 793.1154, indicating substantial price fluctuations during the sample period. The maximum and minimum values of 5807.7200 and 2086.9700 further reflect large cyclical expansions and contractions in the Chinese equity market.

Table 1. Descriptive Statistics of the Closing Prices and log returns of the CSI 300 Index.

Statistics	Closing Price	Log Returns
Mean	3533.5693	0.0000
Standard deviations	793.1154	0.0137
Median	3535.3000	0.0002
Maximum	5807.7200	0.0814
Minimum	2086.9700	−0.0915
Skewness	0.1785	−0.5756
Kurtosis	−0.4688	5.4143
Ljung-Box (Lag = 10)	37,762.7659 (0.0000)	32.0361 (0.0000)
ADF	−1.9345 (0.1168)	−11.198459 (0.0000)

Note: The value in parentheses represents the Minimum Bayes Factor (MBF). Evidence strength is assessed using the minimum Bayes factor (MBF). Smaller values indicate stronger evidence against the null hypothesis (H_0). MBF values are interpreted as follows: <0.01 (decisive), 0.01–0.05 (strong), 0.05–0.10 (moderate), and >0.10 (weak evidence) [48,49].

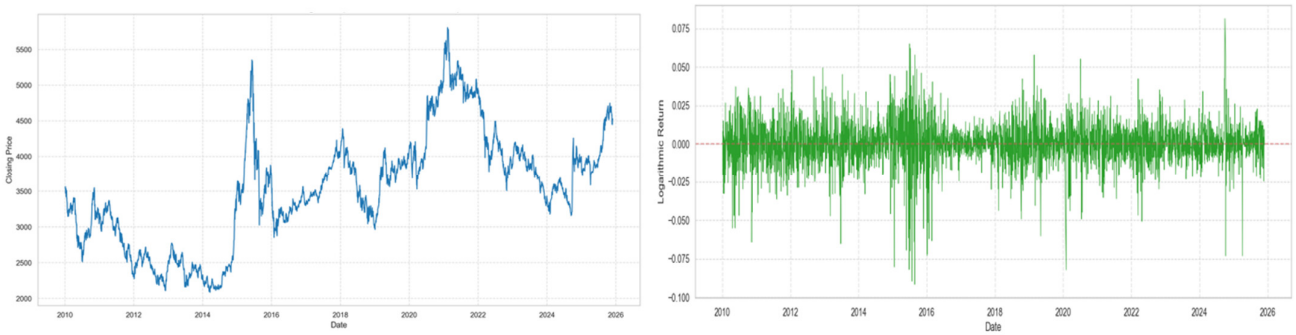
The stationarity results indicate that the raw price series contains a unit root. The ADF statistic for closing prices is −1.9345 (MBF = 0.1168), providing weak evidence against the null hypothesis of non-stationarity. In contrast, the log-return series is strongly stationary, with an ADF statistic of −11.1985 (MBF = 0.0000). These results support the use of log returns for subsequent regime identification and forecasting analysis.

The Ljung–Box statistics indicate significant serial dependence in both price and return series. For log returns, the Ljung–Box statistic at lag 10 equals 32.0361 (MBF = 0.0000), suggesting that short-term temporal dependence remains present even after return transformation. This finding provides empirical motivation for nonlinear and regime-aware forecasting models designed to capture evolving market dynamics.

The distributional statistics additionally reveal non-normal features in the return series. Log returns exhibit negative skewness (−0.5756), indicating greater downside asymmetry, while the kurtosis value of 5.4143 suggests excess tail risk relative to the Gaussian benchmark. These characteristics are consistent with the well-documented empirical properties of financial returns, including asymmetry, volatility clustering, and fat-tailed behavior.

Figure 2 illustrates the daily closing prices and logarithmic returns of the CSI 300 Index over the sample period. Panel (a) shows that the raw closing-price series exhibits substantial trend variation and large cyclical fluctuations, including the pronounced market expansion and correction around 2015, the strong upward movement during 2020–2021,

and renewed volatility after 2022. These characteristics suggest that the price series is non-stationary and influenced by changing market conditions over time.



(**a**) (**b**)

Figure 2. (**a**) CSI 300 Index's Closing Price Plot; (**b**) CSI 300 Index's Log Returns Time Series Plot.

Panel (b) presents the logarithmic return series. Although returns fluctuate around a relatively stable mean near zero, the series exhibits clear volatility clustering and several extreme positive and negative shocks, particularly during 2015–2016 and 2024–2025. These features indicate that the transformed return series remains characterized by time-varying volatility, tail risk, and short-term dependence, which are common empirical properties of financial returns.

Overall, the preliminary data analysis supports three important modeling considerations. First, the non-stationarity of price levels and the stationarity of log returns justify transforming prices into logarithmic returns before model estimation. Second, the presence of serial dependence supports the use of sequential forecasting models rather than purely static specifications. Third, the observed volatility clustering and excess tail behavior motivate the use of regime-aware forecasting models capable of capturing evolving market conditions and nonlinear temporal dynamics.

To evaluate the generalizability of the proposed framework, additional cross-market validation is conducted using the S&P 500 Index (United States) and the Nikkei 225 Index (Japan). These markets provide complementary market environments with different institutional structures, volatility characteristics, and information dynamics, allowing the robustness of the proposed forecasting framework to be assessed across heterogeneous equity markets.

3.2. Regime Identification via Markov Switching

Financial markets commonly exhibit nonlinear dynamics, volatility clustering, and recurrent shifts between adverse, intermediate, and favorable market conditions. To capture these latent states, this study applies a three-regime Markov Switching (MS) model to the Kalman-filtered logarithmic returns of the CSI 300 Index [50,51]. The Kalman Filter first reduces short-term market noise, while the MS model identifies time-varying return–volatility regimes based on the conditional distribution of filtered returns. The estimated regime probabilities are subsequently incorporated into the forecasting framework as regime-aware inputs.

Table 2 reports the estimated regime diagnostics. Regime 0, interpreted as the adverse return–volatility state, contains 1025 observations (26.38% of the sample) and is characterized by a negative mean return (−0.0092) and the highest volatility (0.0138). The minimum return in this regime reaches −0.0915, indicating periods of substantial downside risk and market stress. Regime 1, representing the intermediate market state, contains 1482 observations (38.15% of the sample). This regime exhibits near-zero mean returns (−0.0012) and

the lowest volatility (0.0096), reflecting relatively stable market conditions with limited directional movement. Regime 2, interpreted as the favorable return–volatility state, contains 1378 observations (35.47% of the sample) and is associated with positive average returns (0.0083) and moderate volatility (0.0124). The maximum return of 0.0814 further suggests that this regime captures periods of strong market recovery and upward momentum.

Table 2. Regime diagnostics for the CSI 300 Index.

Regime	Interpretation	Observation	Share	Mean Return	Volatility	Minimum Return	Maximum Return
0	Adverse return-volatility state	1025	0.2638	−0.0092	0.0138	−0.0915	0.0148
1	Intermediate market state	1482	0.3815	−0.0012	0.0096	−0.0694	0.0318
2	Favorable return-volatility state	1378	0.3547	0.0083	0.0124	−0.0731	0.0814

The transition probability matrix reported in Table 3 further confirms that the estimated regimes are economically meaningful and moderately persistent. The diagonal elements of the matrix indicate relatively high probabilities of remaining in the same regime from one trading day to the next. In particular, the favorable return–volatility regime (Regime 2) exhibits the highest persistence, with a self-transition probability of 0.7939, implying an expected duration of approximately 4.85 trading days. The adverse return–volatility regime (Regime 0) also shows substantial persistence, with a probability of 0.7639 and an expected duration of 4.24 trading days. By comparison, the intermediate market state (Regime 1) is less persistent, with a self-transition probability of 0.6962 and an expected duration of 3.29 trading days.

Table 3. Estimated transition probability matrix for the CSI 300 regime-switching model.

From/To	Regime 0 (Adverse Return-Volatility State)	Regime 1 (Intermediate Market State)	Regime 2 (Favorable-Return-Volatility State)	Expected Duration
Regime 0 (Adverse return-volatility State)	0.7639	0.1659	0.0702	4.24 days
Regime 1 (Intermediate market state)	0.1607	0.6962	0.1431	3.29 days
Regime 2 (Favorable return-volatility state)	0.0022	0.2039	0.7939	4.85 days

The off-diagonal transition probabilities indicate gradual adjustment across market states. The probability of moving directly from the adverse to the favorable regime is relatively low (0.0702), while transitions from the adverse to the intermediate regime occur more frequently (0.1659). Likewise, direct transitions from the favorable to the adverse regime are rare (0.0022), whereas transitions to the intermediate regime are more common (0.2039). These patterns suggest that shifts in market conditions typically evolve progressively through intermediate states rather than through abrupt reversals between extreme market regimes. These persistence patterns are further supported by the empirical spell-length distributions reported in Appendix E (Tables A10–A12).

The empirical regime classifications also exhibit economically meaningful spell persistence rather than isolated one-day fluctuations. Favorable and adverse return–volatility regimes tend to cluster over multiple consecutive trading days, while the intermediate regime shows relatively shorter spell durations and more frequent transitions. This pattern further supports the interpretation that the estimated latent states capture persistent market environments rather than purely transitory noise.

Figure 3 presents the Kalman-filtered price series and the corresponding Markov-switching regime classification for the CSI 300 Index. The upper panel compares the raw closing-price series with the Kalman-filtered smoothed series. The two series move closely

together, indicating that the Kalman Filter preserves the broad market trend while reducing short-term observational noise. The vertical dashed line denotes the chronological 80:20 train–test split used for out-of-sample evaluation.

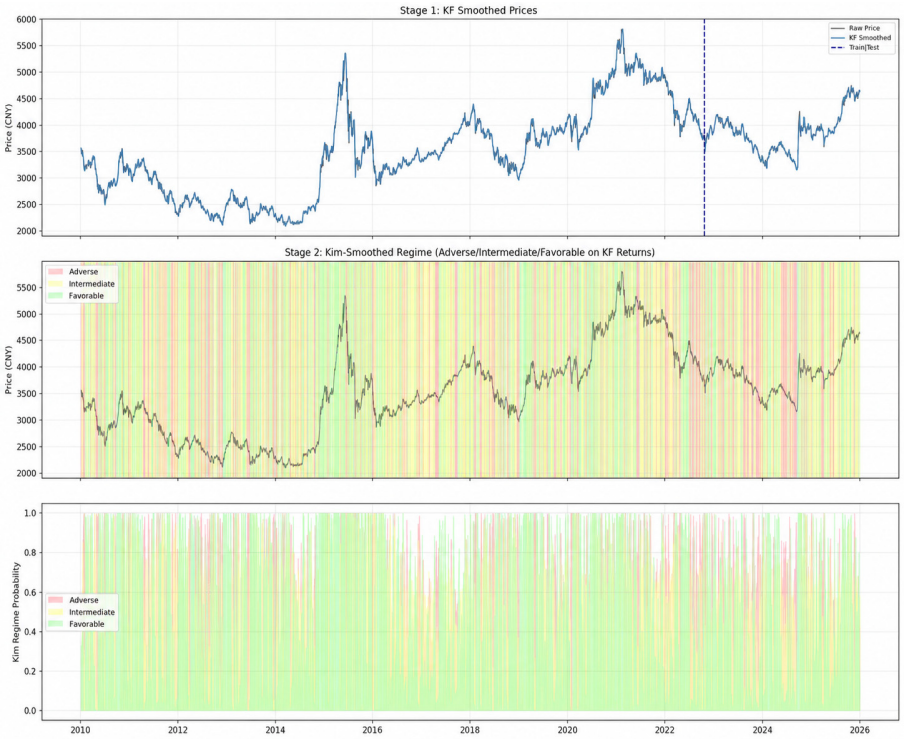


Figure 3. Kalman Filtering and Regime Identification for the CSI 300 Index.

The middle panel reports the estimated dominant regime over time, where green, yellow, and red backgrounds denote favorable return-volatility, intermediate market, and adverse return-volatility states, respectively. The lower panel presents the corresponding regime probabilities, showing that the model frequently assigns high probability to a dominant latent state at each point in time.

The estimated regimes are broadly consistent with major market episodes in the CSI 300 Index. Adverse return-volatility states become more pronounced during the 2015–2016 market correction, the 2018–2019 trade-tension period, and the early stage of the COVID-19 shock in 2020. In contrast, favorable return-volatility states are more evident during strong upward market phases, particularly around 2014–2015 and 2020–2021. Intermediate market states generally appear during relatively stable transition periods without sustained directional movement.

Overall, the results suggest that the Markov Switching model captures economically meaningful changes in market conditions rather than random short-term fluctuations. The estimated regime probabilities and regime-weighted signals are subsequently incorporated into the recurrent-learning framework, allowing the proposed forecasting model to condition predictions on evolving market environments.

3.3. Forecasting Accuracy and Ablation Results: CSI 300

This subsection evaluates the out-of-sample forecasting performance of the proposed framework for the CSI 300 Index. All models are estimated using the first 80% of observations and evaluated on the remaining 20% under a strictly chronological forecasting design. Forecast accuracy is assessed using RMSE, MAE, Theil's inequality coefficient (TIC), mean directional accuracy (MDA), and R^2.

Table 4 reports the forecasting performance of all benchmark and hybrid models. The results show that the proposed KF–MS–LSTM achieves the lowest RMSE (34.7702), the lowest MAE (21.8597), the lowest TIC (0.0045), and the highest R^2 (0.9903). The closely related KF–MS–GRU model also performs strongly, with RMSE and MAE values only slightly above those of KF–MS–LSTM. These results indicate that the proposed regime-aware hybrid framework provides substantial improvements in forecasting accuracy.

Table 4. Out-of-Sample Forecasting Performance for the CSI 300 index under different benchmarks.

Model	RMSE	MAE	TIC	MDA	R^2
ARIMA (0, 0, 0)	42.0690	28.6315	0.0054	1.0000	0.9858
GARCH	42.0496	28.6378	0.0054	0.4839	0.9858
Standard MS	36.2294	24.9813	0.0047	0.6782	0.9895
Pure LSTM	42.5628	28.7112	0.0055	0.4659	0.9855
GARCH-LSTM	43.0640	30.0963	0.0056	0.4929	0.9851
MS-LSTM	35.8860	22.4974	0.0046	0.7349	0.9897
KF-MS-LSTM	**34.7702**	**21.8597**	**0.0045**	**0.7362**	**0.9903**
Pure GRU	42.6103	28.9024	0.0055	0.4852	0.9854
MS-GRU	36.7238	22.7375	0.0047	0.7233	0.9892
KF-MS-GRU	35.0394	21.9059	0.0045	0.7246	0.9901
Transformer	42.0667	28.6321	0.0054	0.4968	0.9858
KF-MS-Transformer	38.3033	26.1022	0.0050	0.6268	0.9882

Note: Bold row denotes the proposed KF–MS–LSTM model. Because ARIMA forecasts are close to zero, its MDA should be interpreted jointly with RMSE, MAE, TIC, R^2, and economic-performance metrics. Repeated-initialization robustness results are reported in Appendix D (Table A9).

A clear distinction emerges between models with and without regime information. ARIMA, GARCH, Pure LSTM, Pure GRU, and Transformer models produce relatively similar RMSE values, ranging from approximately 42.0 to 42.6. Although ARIMA reports an MDA of 1.0000, this value should be interpreted cautiously because near-zero forecasts can mechanically inflate directional-accuracy measures. Therefore, the perfect MDA does not indicate genuine directional forecasting superiority. This is confirmed by ARIMA's weaker magnitude accuracy, with an RMSE of 42.0690, MAE of 28.6315, TIC of 0.0054, and R^2 of 0.9958, all of which are worse than the main regime-aware models. For this reason, the model comparison should rely on the joint evidence from RMSE, MAE, TIC, MDA, and R^2 rather than MDA alone. GARCH–LSTM also performs worse than Pure LSTM in magnitude accuracy, suggesting that volatility information alone does not necessarily improve recurrent-learning forecasts. Similarly, the standard Transformer does not outperform the conventional baselines.

By contrast, models incorporating Markov-switching regime information achieve substantially better performance. Standard MS reduces RMSE to 36.2294, while MS–LSTM further reduces RMSE to 35.8860 and increases MDA to 0.7349. The KF–MS recurrent variants provide the strongest overall results. These findings suggest that regime conditioning, rather than neural-network complexity alone, is the primary source of forecasting improvement in the CSI 300 market.

The ablation analysis further clarifies the contribution of each model component. Within the LSTM family, moving from Pure LSTM to MS–LSTM reduces RMSE from 42.5628 to 35.8860 and improves MDA from 0.4659 to 0.7349, demonstrating the importance of regime-dependent information. Adding the full KF–MS structure further improves forecasting accuracy, although the incremental gain is smaller than that obtained from introducing regime conditioning itself. A similar pattern is observed for the GRU vari-

ants, indicating that the empirical advantage of the framework is robust across recurrent-learning architectures.

Overall, the results indicate that latent regime information substantially improves both magnitude and directional forecasting accuracy, while Kalman filtering provides additional refinement through noise reduction and regime-weighted residual learning. The empirical evidence, therefore, supports the use of a regime-aware hybrid forecasting framework for the CSI 300 Index.

Figure 4 presents the out-of-sample forecasting and trading performance of all benchmark models for the CSI 300 Index over the test period from September 2022 to January 2026. The upper panel compares the actual closing-price series with the corresponding one-step-ahead forecasts. Most models capture the broad market trend, including the decline during 2023–2024, the sharp rebound around September 2024, and the subsequent upward movement. However, regime-conditioned models—particularly KF–MS–LSTM, KF–MS–GRU, MS–LSTM, and MS–GRU—track major turning points and abrupt price adjustments more closely than models based solely on raw return dynamics. This visual evidence is consistent with the statistical results in Table 4, where regime-aware specifications achieve lower RMSE, MAE, and TIC values.

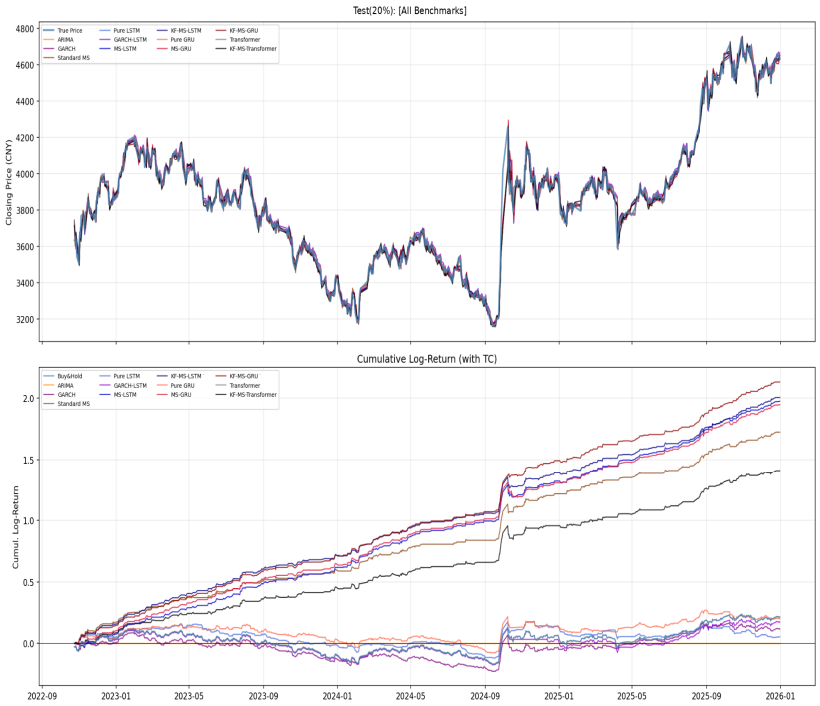


Figure 4. Test-Set Price Forecasting and Cumulative Trading Returns in the CSI 300 Index.

The lower panel reports cumulative log-returns generated by the directional trading strategies after transaction costs. Compared with the forecasting plot, the trading-performance results provide clearer separation across models. Regime-conditioned models consistently outperform the non-regime benchmarks throughout most of the evaluation period. KF–MS–GRU achieves the highest cumulative return, followed closely by KF–MS–LSTM and MS–LSTM. In contrast, ARIMA, GARCH, Pure LSTM, Pure GRU, GARCH–LSTM, and the standalone Transformer generate relatively weak trading performance despite tracking the general price trend. The Buy-and-Hold benchmark also underperforms the leading regime-aware models, suggesting that the proposed framework extracts useful market-timing information beyond passive exposure to the overall market trend.

Overall, Figure 4 reinforces the main empirical findings of the study. Regime-aware models improve both forecasting accuracy and economic trading performance, while

Kalman filtering and recurrent nonlinear learning provide additional refinement to the regime-conditioned forecasting structure. The results also indicate that the empirical advantage of the framework is robust across both LSTM- and GRU-based recurrent architectures.

3.4. Statistical Comparison Using Bootstrap DM Tests

To statistically compare forecasting accuracy across competing models, this study applies the Bootstrap Diebold–Mariano (DM) test to assess the robustness of model prediction performance [52]. The DM test, originally proposed by Francis X. Diebold [53], evaluates the null hypothesis that two competing forecasting models possess equal predictive accuracy. It therefore provides a formal statistical framework for determining whether differences in out-of-sample forecast performance are statistically significant.

Let d_t denote the loss differential between two competing forecasting models at time t. Under the null hypothesis, the two models have equal predictive accuracy: $H_0 : E(d_t) = 0$. The corresponding DM statistic is computed as

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})}}, \tag{38}$$

where $\bar{d}$ denotes the average loss differential and $\hat{V}(\bar{d})$ represents its estimated variance.

Following standard forecasting practice, the mean squared error (MSE) is used as the loss function. Because financial-return forecast errors commonly exhibit serial dependence and non-normality, this study employs the Bootstrap Diebold–Mariano test with 1000 bootstrap replications to improve inference robustness. In addition to conventional significance testing, evidence strength is evaluated using the Modified Bayes Factor (MBF) classification framework [48,49]. Smaller MBF values indicate stronger evidence against the null hypothesis of equal predictive accuracy.

Table 5 reports the bootstrap Diebold–Mariano (DM) test results comparing KF–MS–LSTM with all benchmark models using squared-error loss and 1000 bootstrap replications. Statistical evidence is evaluated using the Modified Bayes Factor (MBF), where smaller values indicate stronger evidence against the null hypothesis of equal predictive accuracy. Negative t-statistics indicate that KF–MS–LSTM produces a lower average forecast loss than the competing specification.

Table 5. Bootstrap Diebold–Mariano Test Results for KF–MS–LSTM Relative to Benchmark Models.

Model A	Model B	t-Statistic	MBF	Evidence	Better Model	95% CI
KF-MS-LSTM	ARIMA	−7.5664	0.0000	Decisive	KF-MS-LSTM	[−712.6510, −414.4090]
	GARCH	−7.7034	0.0000	Decisive	KF-MS-LSTM	[−709.7573, −415.5617]
	Standard MS	−1.8010	0.5324	Weak	KF-MS-LSTM	[−212.7203, 16.4355]
	Pure LSTM	−7.1494	0.0000	Decisive	KF-MS-LSTM	[−766.9097, −447.1168]
	GARCH-LSTM	−8.3651	0.0000	Decisive	KF-MS-LSTM	[−801.7879, −501.4301]
	MS-LSTM	−1.3869	0.7661	Weak	KF-MS-LSTM	[−194.6633, 29.2103]
	Pure GRU	−7.7820	0.0000	Decisive	KF-MS-LSTM	[−752.0204, −457.2309]
	MS-GRU	−1.5384	0.6399	Weak	KF-MS-LSTM	[−340.4631, 25.2441]
	KF-MS-GRU	−0.5583	1.0000	Weak	KF-MS-LSTM	[−94.1830, 40.7991]
	Transformer	−7.5662	0.0000	Decisive	KF-MS-LSTM	[−712.4032, −414.0288]
	KF-MS-Transformer	−5.3386	0.0000	Decisive	KF-MS-LSTM	[−349.9403, −156.5648]

Note: Smaller MBF values indicate stronger evidence against the null hypothesis of equal predictive accuracy. MBF classifications follow [49]: <0.01 (decisive), 0.01–0.05 (strong), 0.05–0.10 (moderate), and >0.10 (weak evidence).

The results provide decisive evidence that KF–MS–LSTM outperforms ARIMA, GARCH, Pure LSTM, GARCH–LSTM, Pure GRU, Transformer, and KF–MS–Transformer. In these comparisons, the t-statistics are strongly negative, the MBF values are close to

zero, and the 95% confidence intervals lie entirely below zero. These findings confirm that the lower RMSE, MAE, and TIC values reported in Table 4 are statistically robust relative to conventional econometric, volatility-based, recurrent-learning, and attention-based benchmark models.

The largest performance differences are observed against GARCH–LSTM, Pure GRU, Pure LSTM, and Transformer models. For example, KF–MS–LSTM decisively outperforms GARCH–LSTM with a t-statistic of -8.3651 and a 95% confidence interval of $[-801.7879, -501.4301]$. These results suggest that models relying only on volatility inputs or raw return dynamics are insufficient for capturing the nonlinear and regime-dependent structure of the CSI 300 Index. Similarly, the weak performance of the standalone Transformer indicates that attention-based learning alone does not necessarily improve forecasting accuracy without informative regime-conditioned inputs.

By contrast, the evidence becomes weaker when KF–MS–LSTM is compared with other regime-conditioned models, including Standard MS, MS–LSTM, MS–GRU, and KF–MS–GRU. Although the corresponding t-statistics remain negative, the MBF values indicate weak evidence, and the 95% confidence intervals include zero. In particular, the comparisons with MS–LSTM and KF–MS–GRU suggest that once regime conditioning is incorporated, the incremental gains from Kalman filtering and recurrent-architecture choice become more moderate.

3.5. Economic Value and Trading Tests

Statistical forecasting accuracy alone does not necessarily imply economic usefulness in financial markets. Therefore, this subsection evaluates whether the forecasting models generate economically meaningful trading performance for the CSI 300 Index. As demonstrated below, the practical value of a forecasting model ultimately depends on its ability to support profitable investment decisions rather than merely minimizing statistical forecast errors.

Following the economic evaluation framework introduced in Section 2.11.2, a directional trading simulation is implemented using one-step-ahead forecasts. Specifically, when the predicted closing price for day $t+1$ exceeds the current observed price $(f_{t+1} > y_t)$, a long (buy) position is taken; otherwise, the position is closed or held in cash $(f_{t+1} < y_t)$ [54]. The cumulative trading return is computed as

$$R = 100 \times \left(\sum_{t=1}^{b} \frac{f_{t+1} - y_t}{y_t} - B + \sum_{t=1}^{s} \frac{y_t - f_{t+1}}{y_t} - C \right), \tag{39}$$

where b and s denote the number of buy and sell trading periods, respectively, while B and C represent transaction costs associated with trading activity. To reflect realistic trading conditions in the Chinese stock market, transaction costs are set at 0.1% per transaction [54].

The economic evaluation considers cumulative return, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, win rate, turnover, Calmar ratio, and information ratio. In addition, transaction-cost sensitivity analysis is conducted to assess whether the reported trading performance remains robust under alternative cost assumptions. The results presented below examine whether improvements in statistical forecasting accuracy translate into superior directional trading performance and economically meaningful portfolio outcomes across competing forecasting models.

Table 6 reports the economic performance of the out-of-sample trading strategies for the CSI 300 Index. Overall, models incorporating regime information substantially outperform non-regime benchmarks in cumulative return, risk-adjusted return, and drawdown control. The results indicate that regime conditioning contributes more to trading performance than neural-network complexity alone.

Table 6. Economic Value of Trading Strategies under All Benchmarks.

Model A	Cumulative Return (%)	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio	MDD	Win Rate	Turnover	Calmar	Info Ratio
ARIMA	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	−0.3997
GARCH	−0.1115	−0.0376	0.1509	−0.2491	−0.3577	0.3900	0.3012	−0.1051	−1.2736
Standard MS	4.0967	0.6959	0.1293	5.3828	−0.0705	0.3076	0.1236	9.8683	4.4399
Pure LSTM	−0.1916	−0.0667	0.1373	−0.4853	−0.3278	0.2111	0.3423	−0.2033	−1.3052
GARCH-LSTM	0.1776	0.0545	0.1639	0.3322	−0.2480	0.4517	0.0103	0.2196	−0.2944
MS-LSTM	4.6386	0.7524	0.1297	5.8007	−0.0949	0.3616	0.3166	7.9267	4.8344
KF-MS-LSTM	4.9559	0.7837	0.1300	6.0276	−0.0867	0.3552	0.2857	9.0416	5.0641
Pure GRU	−0.0857	−0.0286	0.1343	−0.2132	−0.3135	0.2831	0.3719	−0.0913	−0.8973
MS-GRU	4.5395	0.7423	0.1311	5.6629	−0.1182	0.3604	0.3037	6.2797	4.8462
KF-MS-GRU	5.5419	0.8389	0.1206	6.9573	−0.0277	0.3243	0.3269	30.3170	4.9115
Transformer	0.2370	0.0714	0.1726	0.4138	−0.2480	0.4968	0.0000	0.2879	0.0000
KF-MS-Transformer	2.5045	0.5019	0.1281	3.9186	−0.1100	0.2793	0.2008	4.5613	3.0724
Buy & Hold	0.2370	0.0714	0.1726	0.4138	−0.2480	0.4968	1.0000	0.2879	0.0000

Note: TC = 0.1% one-way transaction cost. Strategy: long position when the predicted price is greater than the actual price, flat otherwise. MDD = maximum peak-to-trough drawdown during the test period.

Among all specifications, KF–MS–GRU achieves the strongest economic performance, with the highest cumulative return (554.19%), annualized return (83.89%), Sharpe ratio (6.9573), and the lowest maximum drawdown (−2.77%). The proposed KF–MS–LSTM also performs strongly, generating a cumulative return of 495.59%, an annualized return of 78.37%, and a Sharpe ratio of 6.0276. MS-LSTM, MS-GRU, and Standard MS likewise produce substantially higher returns and Sharpe ratios than conventional econometric and standalone deep-learning benchmarks. These findings suggest that latent regime information provides economically valuable market-timing signals.

In contrast, non-regime models exhibit considerably weaker trading performance. ARIMA generates no trading return because its forecasts remain close to zero under the selected specification. GARCH, Pure LSTM, and Pure GRU produce negative cumulative returns and negative Sharpe ratios, indicating that statistical forecasting alone does not necessarily translate into profitable trading signals. GARCH-LSTM and the standalone Transformer perform slightly better, but their economic value remains limited relative to regime-conditioned models. Importantly, KF–MS–Transformer substantially improves upon the standalone Transformer, further supporting the conclusion that regime information enhances both recurrent and attention-based forecasting architectures.

The comparison between Standard MS, MS-LSTM, KF–MS–LSTM, and KF–MS–GRU also provides insight into the incremental contribution of each modeling component. Standard MS already delivers a strong economic performance, indicating that regime-dependent mean dynamics contain substantial predictive value. Adding recurrent nonlinear learning through MS-LSTM and MS-GRU further improves cumulative return and Sharpe performance, while the KF–MS specifications provide additional refinement through noise reduction and regime-aware feature construction. However, the relatively close performance among the leading regime-conditioned models suggests that the primary source of economic improvement originates from regime identification rather than from a specific neural-network architecture.

Overall, Table 6 complements the statistical results in Tables 4 and 5. The findings show that lower forecast errors and stronger directional accuracy can translate into economically meaningful gains when the forecasting framework incorporates latent market-state information. At the same time, the results should be interpreted comparatively rather than as evidence of fully implementable institutional trading performance, because the simulation abstracts from additional real-world frictions such as slippage, liquidity constraints, taxes, and execution delays.

3.6. Robustness Check

To further evaluate the stability and practical reliability of the proposed framework, this subsection conducts four robustness checks. First, transaction-cost sensitivity analysis examines whether the reported trading profitability remains robust under alternative trading frictions. Second, tail-exclusion diagnostics assess whether the economic performance is driven primarily by a small number of extreme observations. Third, lookback-window sensitivity tests whether the forecasting performance depends heavily on a specific sequence length. Finally, Sharpe-stability diagnostics evaluate whether the reported risk-adjusted performance is robust to sampling variation and isolated trading days. Full results are reported in Appendix B.

The transaction-cost sensitivity analysis shows that the regime-aware models remain economically profitable under low-to-moderate trading frictions. As reported in Appendix B Table A3, KF–MS–LSTM maintains positive annual returns of 0.8491, 0.6598, and 0.3374 when one-way transaction costs increase from 0.05% to 0.20% and 0.50%, respectively, while its Sharpe ratio declines from 6.5477 to 5.0367 and 2.4774. KF–MS–GRU exhibits similar robustness, recording annual returns of 0.9162, 0.6935, and 0.3227 under the same cost assumptions. MS-LSTM and MS-GRU also remain profitable at the 0.50% transaction-cost level. In contrast, non-regime models such as GARCH, Pure LSTM, and Pure GRU remain weak or become negative as trading costs rise. When transaction costs increase to the extreme level of 1.00%, most neural regime-aware models become unprofitable, whereas Standard MS remains positive with an annual return of 0.2814 and a Sharpe ratio of 2.1269, likely because of its substantially lower turnover. These results suggest that regime-conditioned signals retain economic value under realistic implementation costs, although profitability remains sensitive to turnover and execution frictions.

The tail-exclusion diagnostics further indicate that the profitability of regime-aware models is not driven solely by a few extreme market observations. Appendix B Table A4 shows that KF–MS–LSTM continues to generate positive annual returns of 0.6952, 0.5430, and 0.4097 after excluding 1%, 5%, and 10% from each tail of the return distribution, respectively, while its Sharpe ratios remain high at 7.3609, 8.1290, and 8.3649. KF–MS–GRU and MS-LSTM exhibit similar stability across all exclusion thresholds, whereas non-regime models remain substantially weaker and often produce negative returns under stricter exclusions. These findings suggest that the economic value of the regime-aware framework does not depend solely on a small number of extreme-return days.

The lookback-window sensitivity analysis, reported in Appendix B Table A5, shows that the proposed framework is not dependent on a single sequence-length specification. Across lookback windows of 5, 10, and 20 trading days, KF–MS–LSTM maintains relatively stable forecasting accuracy, with RMSE values ranging from 34.7702 to 36.1439 and Sharpe ratios remaining above 5.0. Shorter windows perform slightly better, suggesting that recent regime-conditioned information is particularly informative for one-step-ahead forecasting.

Additional Sharpe-stability diagnostics are reported in Appendix B Table A6. A block-bootstrap procedure is used to estimate Sharpe-ratio confidence intervals while preserving short-run dependence in strategy returns, and leave-one-out diagnostics test whether performance is driven by individual trading days. For KF–MS–LSTM, the baseline Sharpe ratio is 6.0276, while the bootstrap mean Sharpe ratio is 6.1896 with a 95% confidence interval of [4.4557, 7.9912]. The leave-one-out Sharpe range remains narrow and positive, ranging from 5.9386 to 6.7431. Furthermore, after excluding the top 1% of positive-return days, KF–MS–LSTM still records a Sharpe ratio of 5.5317, while KF–MS–GRU remains at 6.7348. These results indicate that the strong economic performance of the regime-aware models reflects persistent timing ability rather than dependence on a few isolated observations.

To sum up, the robustness checks support the stability of the proposed regime-aware forecasting framework across alternative transaction-cost assumptions, tail-event exclusions, lookback-window specifications, and Sharpe-stability diagnostics. The evidence consistently indicates that regime information is the primary source of forecasting and trading improvement, while Kalman filtering and nonlinear learning provide additional refinement to the regime-conditioned signal.

3.7. Cross-Market Validation on S&P 500 and Nikkei 225

The CSI 300 results suggest that regime-conditioned models improve both forecasting accuracy and economic performance, primarily through the incorporation of latent market-state information. To evaluate whether these findings generalize beyond the Chinese market, the empirical analysis is extended to two additional equity indices: the S&P 500 in the United States and the Nikkei 225 in Japan. These markets differ substantially in market maturity, institutional structure, investor composition, and volatility dynamics, providing a useful setting for cross-market validation of the proposed framework.

3.7.1. Data Characteristics and Regime Identification

Table 7 reports the descriptive statistics for the logarithmic returns of the S&P 500 and Nikkei 225 indices. Both return series exhibit small positive mean returns and negative skewness, indicating asymmetric downside risk. The Nikkei 225 shows slightly higher return volatility, with a standard deviation of 0.0134 compared with 0.0109 for the S&P 500. Both markets also display substantial excess kurtosis, particularly the S&P 500, which records a kurtosis value of 16.5724. These results confirm that both return distributions are non-normal and fat-tailed, consistent with the stylized characteristics of financial time series.

Table 7. Descriptive Statistics of the Log Returns of the S&P 500 and Nikkei 225.

Statistics	S&P 500	Nikkei 225
Mean	0.0005	0.0004
Standard deviations	0.0109	0.0134
Median	0.0007	0.0007
Maximum	0.0909	0.0974
Minimum	−0.1277	−0.1323
Skewness	−0.6150	−0.4896
Kurtosis	16.5724	10.7124
Ljung-Box (Lag = 10)	238.1911 (0.0000)	24.3763 (0.090748)
ADF	−13.7242 (0.0000)	−36.4060 (0.0000)

Note: The value in parentheses represents the Minimum Bayes Factor (MBF). MBF values are interpreted as follows: <0.01 (decisive), 0.01–0.05 (strong), 0.05–0.10 (moderate), and >0.10 (weak evidence) [49].

The stationarity and serial-dependence diagnostics further support the use of dynamic forecasting models. The ADF statistics are −13.7242 for the S&P 500 and −36.4060 for the Nikkei 225, providing decisive evidence against the unit-root null and confirming that both log-return series are stationary. The Ljung–Box statistic for the S&P 500 is 238.1911 (MBF = 0.0000), indicating strong serial dependence. For the Nikkei 225, the Ljung–Box statistic is 24.3763 with weaker MBF evidence (0.090748), suggesting that short-term dependence remains present but less pronounced than in the U.S. market. Overall, the descriptive statistics indicate that both validation markets exhibit volatility clustering, tail risk, asymmetry, and short-run dependence, although with differing degrees of persistence and volatility intensity. These characteristics provide empirical support for applying regime-aware and sequential forecasting models across heterogeneous international equity markets.

Table 8 reports the three-state Markov Switching regime diagnostics for the S&P 500 and Nikkei 225. In both markets, the estimated regimes can be interpreted as adverse return-volatility, intermediate market, and favorable return-volatility states, confirming that the three-state structure remains economically meaningful beyond the CSI 300 market.

Table 8. Regime diagnostics for the S&P 500 and Nikkei 225.

Stock Market	Regime	Interpretation	Observations	Share	Mean Return	Volatility	Minimum Return	Maximum Return
S&P 500	0	Adverse return-volatility state	739	0.1834	−0.0091	0.0123	−0.1277	0.0161
Nikkei 225			1057	0.2704	−0.0098	0.0144	−0.1323	0.0552
S&P 500	1	Intermediate market state	1376	0.3414	−0.0012	0.0091	−0.0532	0.0888
Nikkei 225			1930	0.4937	0.0017	0.0102	−0.0462	0.0974
S&P 500	2	Favorable return-volatility state	1915	0.4752	0.0054	0.0085	−0.0352	0.0909
Nikkei 225			922	0.2359	0.0094	0.0101	−0.0179	0.0874

For the S&P 500, Regime 0 captures adverse market conditions with a negative mean return (−0.0091) and relatively high volatility (0.0123), while Regime 2 represents favorable market conditions with a positive mean return of 0.0054 and the largest sample share (47.52%). This suggests that the U.S. market spends a substantial proportion of the sample period in positive-return states. Regime 1 corresponds to an intermediate low-volatility state with returns close to zero. The Nikkei 225 exhibits a similar regime structure but with some notable differences. Regime 0 again corresponds to adverse high-volatility conditions, with a mean return of −0.0098 and the highest volatility among the three states. However, the intermediate regime accounts for the largest share of observations (49.37%), while the favorable regime has a smaller sample share than in the S&P 500. These results suggest that the Japanese market experiences more prolonged transitional periods and less persistent directional trends than the U.S. market.

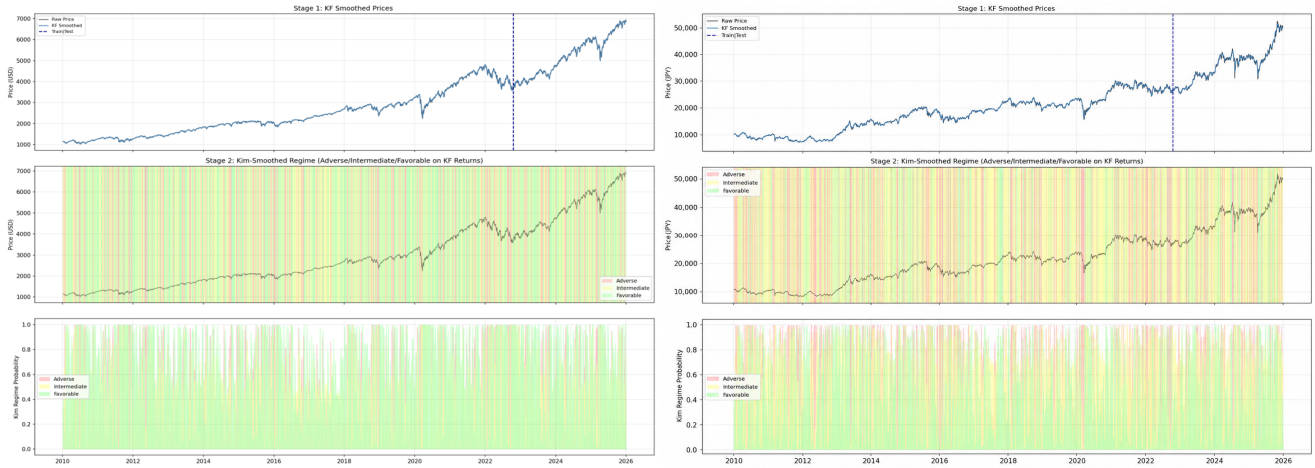
The transition probability matrices for the S&P 500 and Nikkei 225 further confirm that the three-state structure captures short-horizon return-volatility regimes with measurable persistence rather than random daily classifications. As shown in Table 9, for the S&P 500, the favorable return-volatility state is the most persistent, with a self-transition probability of 0.8251 and an expected duration of approximately 5.72 trading days, while the adverse and intermediate states have expected durations of 3.17 and 2.78 trading days, respectively. This suggests that favorable market conditions in the U.S. market tend to persist longer than adverse or intermediate states. For the Nikkei 225, the three regimes display more balanced persistence: the adverse and intermediate states both have expected durations of about 4.13 trading days, while the favorable state lasts approximately 3.96 trading days. This pattern suggests more frequent and symmetric switching across market states in Japan, consistent with the Nikkei 225's more transitional return-volatility structure.

Overall, the regime diagnostics indicate that the proposed Markov Switching framework identifies economically interpretable market states across structurally different international equity markets. Although the relative regime frequencies differ across markets, the common presence of asymmetric returns, volatility heterogeneity, and persistent regime behavior supports the use of a unified regime-aware forecasting framework in the subsequent cross-market validation analysis.

Figure 5 presents the regime-classification results for the S&P 500 and Nikkei 225 indices. In the S&P 500, the favorable return-volatility state dominates a substantial proportion of the sample period, consistent with the long upward trend of the U.S. equity market during 2010–2025. Adverse return-volatility states appear mainly during periods of market stress and short-term corrections, while intermediate states occur during relatively stable transitional periods.

Table 9. Estimated transition probability matrix for the S&P 500 and Nikkei 2225 regime-switching models.

From/To	Regime 0 (Adverse Return-Volatility State)	Regime 1 (Intermediate Market State)	Regime 2 (Favorable-Return-Volatility State)	Expected Duration
S&P 500				
Regime 0 (Adverse return-volatility State)	0.6847	0.2179	0.0974	3.17 days
Regime 1 (Intermediate market state)	0.1687	0.6400	0.1913	2.78 days
Regime 2 (Favorable return-volatility state)	0.0000	0.1749	0.8251	5.72 days
Nikkei 225				
Regime 0 (Adverse return-volatility State)	0.7578	0.2375	0.0047	4.13 days
Regime 1 (Intermediate market state)	0.1239	0.7579	0.1182	4.13 days
Regime 2 (Favorable return-volatility state)	0.0174	0.2354	0.7473	3.96 days



(**a**) (**b**)

Figure 5. (**a**) State Identification of the S&P 500 Index; (**b**) State Identification of the Nikkei 225 Index.

The Nikkei 225 also exhibits economically interpretable adverse, intermediate, and favorable return-volatility states. However, compared with the S&P 500, the Japanese market shows more frequent regime transitions and a larger share of intermediate states, suggesting a less persistent directional trend and greater sensitivity to external shocks and global market conditions.

Overall, Figure 5 confirms that the Markov Switching model identifies economically meaningful latent market states across both validation markets. Although the relative persistence and frequency of regimes differ across countries, the general regime structure remains stable, supporting the cross-market applicability of the proposed regime-aware forecasting framework.

3.7.2. Forecasting Accuracy Across Markets

Table 10 reports the out-of-sample forecasting performance for the S&P 500 and Nikkei 225 using the same benchmark structure and evaluation criteria as in the CSI 300 analysis.

Overall, models incorporating Markov-switching regime information consistently outperform classical econometric, volatility-based, pure deep-learning, and attention-based benchmarks across both markets. In the S&P 500, Standard MS achieves the strongest overall statistical accuracy, recording the lowest RMSE (42.3563), lowest MAE (29.1528), lowest TIC (0.0040), highest MDA (0.7196), and highest R^2 (0.9978). MS-GRU and KF–MS–GRU also perform competitively, with RMSE values of 44.1948 and 44.7074, respectively. KF–MS–LSTM achieves an RMSE of 45.4777 and an MDA of 0.6563, outperforming ARIMA,

GARCH, Pure LSTM, Pure GRU, GARCH-LSTM, Transformer, and KF–MS–Transformer, although remaining slightly weaker than Standard MS in this market. These results suggest that, for the S&P 500, regime information alone already provides a strong forecasting signal, while the incremental contribution of nonlinear recurrent learning is more moderate.

Table 10. Out-of-Sample Forecasting Performance for the S&P 500 and Nikkei 225 under different benchmarks.

Model	RMSE	MAE	TIC	MDA	R^2
S&P 500					
ARIMA (1, 0, 4)	50.5457	35.3247	0.0048	0.5087	0.9969
GARCH	50.3557	34.8124	0.0048	0.5471	0.9970
Standard MS	**42.3563**	**29.1528**	**0.0040**	**0.7196**	**0.9978**
Pure LSTM	49.6560	34.8584	0.0047	0.5124	0.9970
GARCH-LSTM	50.3756	35.1695	0.0048	0.4926	0.9970
MS-LSTM	44.9211	30. 4499	0.0042	0.6526	0.9976
KF-MS-LSTM	45.4777	31.1010	0.0043	0.6563	0.9975
Pure GRU	50. 2694	34. 8951	0.0047	0.5310	0.9970
MS-GRU	44.1948	28.5805	0.0042	0.6911	0.9977
KF-MS-GRU	44.7074	30.4681	0.0042	0.6873	0.9976
Transformer	50.4067	34.8538	0.0048	0.5558	0.9970
KF-MS-Transformer	46.1153	31.6548	0.0044	0.6613	0.9975
Nikkei 225					
ARIMA (0, 0, 0)	510.4329	342.6425	0.0069	1.0000	0.9929
GARCH	508.9343	341.2550	0.0069	0.5486	0.9929
Standard MS	429.3453	283.4326	0.0058	0.6854	0.9950
Pure LSTM	509.5875	342.0966	0.0069	0.5460	0.9929
GARCH-LSTM	510.9095	343.5715	0.0069	0.4668	0.9929
MS-LSTM	443.8478	269.8774	0.0060	0.7161	0.9946
KF-MS-LSTM	439.3253	264.8392	0.0060	0.7442	0.9947
Pure GRU	510.3533	343.8293	0.0069	0.5064	0.9928
MS-GRU	441.4236	266.7494	0.0060	0.7379	0.9947
KF-MS-GRU	**428.7409**	**263.8659**	**0.0058**	**0.7391**	**0.9950**
Transformer	509.9336	341.8209	0.0069	0.5486	0.9929
KF-MS-Transformer	471.1454	308.3979	0.0064	0.6458	0.9939

Note: Bold row indicates best-performing models in most metrics. All metrics competed against actual S&P 500 closing prices.

For Nikkei 225, the regime-aware hybrid models also dominate the non-regime benchmarks. KF–MS–GRU achieves the best overall magnitude forecasting performance, with the lowest RMSE (428.7409), lowest MAE (263.8659), lowest TIC (0.0058), and highest R^2 (0.9950). KF–MS–LSTM performs strongly in directional prediction, recording the highest meaningful MDA of 0.7442 among the non-degenerate forecasting models and the second-lowest MAE of 264.8392. MS-LSTM and MS-GRU also substantially outperform Pure LSTM and Pure GRU, indicating that regime conditioning remains the primary source of forecasting improvement in the Japanese market.

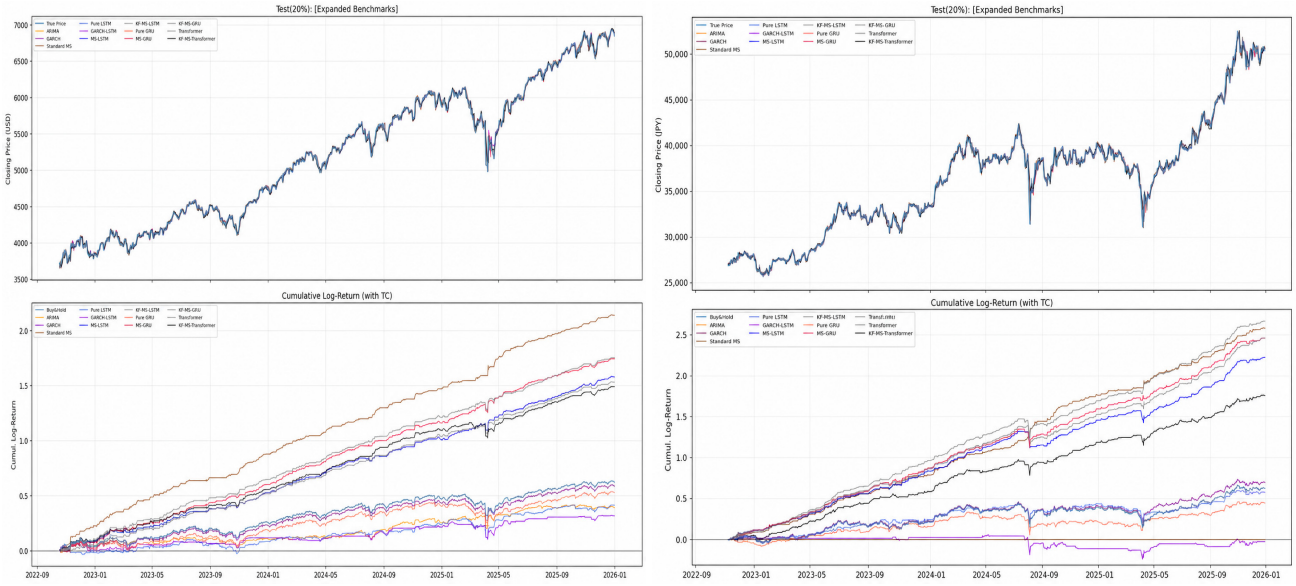
The results also indicate that pure deep-learning and attention-based architectures do not consistently generate superior forecasting performance without regime information. Across both markets, Pure LSTM, Pure GRU, GARCH-LSTM, and the standalone Transformer generally produce larger RMSE and MAE values than the regime-conditioned models. For example, in the Nikkei 225 market, the standalone Transformer records an RMSE of 509.9336 and a TIC of 0.0069, compared with 428.7409 and 0.0058 for KF–MS–GRU. Similarly, in the S&P 500 market, Pure LSTM produces an RMSE of 49.6560, substantially above the 42.3563 achieved by Standard MS. These findings reinforce the conclusion that

regime information contributes more to forecasting improvement than neural-network complexity alone.

Notably, the ARIMA (0, 0, 0) specification for the Nikkei 225 reports an MDA value of 1.0000. However, this result should be interpreted cautiously because the nearly constant forecasts generated by this specification mechanically align with the dominant directional pattern in the test sample. Its weaker magnitude accuracy, including an RMSE of 510.4329, MAE of 342.6425, and TIC of 0.0069, confirms that the perfect MDA does not reflect genuine forecasting superiority. Therefore, model comparisons should rely on the joint evidence from RMSE, MAE, TIC, MDA, and R^2, rather than on directional accuracy alone.

Taken together, the cross-market results support the robustness of the proposed regime-aware forecasting framework rather than the unconditional dominance of a single architecture. Although the best-performing specification varies across markets—Standard MS in the S&P 500 and KF–MS–GRU in the Nikkei 225—models incorporating Markov-switching regime information consistently outperform most non-regime benchmarks. This pattern suggests that regime conditioning is the most stable source of forecasting improvement across heterogeneous market environments, while the incremental contribution of Kalman filtering and the choice between LSTM, GRU, and Transformer architectures remain market dependent.

Figure 6 presents the out-of-sample forecasting and trading performance for the S&P 500 (panel a) and Nikkei 225 (panel b) during the test period. In both markets, the upper panels show that most benchmark models track the broad movement of the actual price series relatively closely, indicating that price-level forecasting differences are modest at the visual level. However, the lower panels reveal substantially clearer differences in economic performance.



(**a**) (**b**)

Figure 6. (**a**) Forecasting Performance of the S&P 500; (**b**) Forecasting Performance of Nikkei 225.

For the S&P 500, Standard MS generates the strongest cumulative trading return, followed by KF–MS–GRU, MS-GRU, KF–MS–LSTM, and MS-LSTM, all of which remain consistently above the Buy-and-Hold benchmark throughout most of the evaluation period. This visual evidence is consistent with Table 10, where Standard MS records the lowest RMSE (42.3563), the lowest TIC (0.0040), and the highest MDA (0.7196). By contrast, ARIMA, GARCH, Pure LSTM, Pure GRU, GARCH-LSTM, and the standalone Transformer

produce substantially weaker cumulative-return trajectories, indicating that models without regime conditioning generate less economically useful trading signals in the U.S. market.

For the Nikkei 225, the lower-right panel shows even stronger separation between regime-aware and non-regime models. KF–MS–GRU and KF–MS–LSTM produce the strongest cumulative trading performance, followed by MS-GRU, MS-LSTM, and Standard MS. These results align with Table 10, where KF–MS–GRU achieves the lowest RMSE (428.7409), the lowest TIC (0.0058), and the highest R^2 (0.9950), while KF–MS–LSTM records the highest meaningful MDA (0.7442). A temporary decline is visible around the 2024 market correction, after which the regime-aware strategies recover rapidly and continue to outperform the non-regime benchmarks. In contrast, ARIMA, GARCH, Pure LSTM, Pure GRU, GARCH-LSTM, and the standalone Transformer remain substantially lower over most of the test horizons.

Overall, Figure 6 provides graphical support for the cross-market forecasting results reported in Table 9. Across both markets, regime-conditioned models consistently deliver stronger economic value than non-regime benchmarks, even when most models appear visually similar in price-level forecasting. The leading specification differs across markets—Standard MS in the S&P 500 and KF–MS–GRU/KF–MS–LSTM in the Nikkei 225—indicating that the benefits of regime conditioning are robust across heterogeneous markets, while the incremental contribution of Kalman filtering and nonlinear recurrent learning remains market dependent.

3.7.3. Statistical and Economic Validation

To further validate the robustness of the proposed regime-aware forecasting framework across international markets, this subsection examines the statistical significance of forecasting improvements using the Bootstrap Diebold–Mariano (DM) test. Table 11 reports the cross-market DM test results for the S&P 500 and Nikkei 225 using KF–MS–LSTM as the reference model. A negative *t*-statistic indicates that KF–MS–LSTM achieves lower forecast loss than the competing benchmark, while the MBF values classify the strength of the statistical evidence.

Table 11. The Bootstrap DM test results on the S&P 500 and Nikkei 225.

Model A	Model B	S&P 500 Index		Nikkei 225 Index	
		***t*-Statistic**	**MBF**	***t*-Statistic**	**MBF**
KF-MS-LSTM	ARIMA	−6.2234	0.0000	−5.8344	0.0000
	GARCH	−5.3340	0.0000	−5.7421	0.0000
	Standard MS	3.2575	0.0720	0.7980	1.0000
	Pure LSTM	−6.2756	0.0000	−5.8686	0.0000
	GARCH-LSTM	−3.7973	0.0000	−5.9096	0.0000
	MS-LSTM	0.8324	1.0000	−0.6834	1.0000
	Pure GRU	−5.7742	0.0000	−6.0466	0.0000
	MS-GRU	1.0808	0.9482	−0.2335	1.0000
	KF-MS-GRU	2.6473	0.1252	1.6647	0.6078
	Transformer	−5.2101	0.0000	−5.8060	0.0000
	KF-MS-Transformer	−0.6361	1.0000	−3.0201	0.0720

Note: MBF values are interpreted as follows: <0.01 (decisive), 0.01–0.05 (strong), 0.05–0.10 (moderate), and >0.10 (weak evidence) [49].

Overall, KF–MS–LSTM decisively outperforms the non-regime and non-hybrid benchmarks in both markets. For the S&P 500, the comparisons against ARIMA, GARCH, Pure LSTM, GARCH-LSTM, Pure GRU, and Transformer all produce strongly negative *t*-statistics ranging from −3.7973 to −6.2756, with MBF values close to zero, indicating decisive evidence in favor of KF–MS–LSTM. A similar pattern appears in the Nikkei 225,

where the corresponding *t*-statistics range from −5.7421 to −6.0466, again accompanied by near-zero MBF values. These results confirm that incorporating regime information and nonlinear learning substantially improve forecasting performance relative to conventional time-series, volatility-based, pure recurrent, and standalone attention-based models.

However, the evidence becomes weaker when KF–MS–LSTM is compared with other regime-aware specifications. In the S&P 500, Standard MS records a positive *t*-statistic of 3.2575 with an MBF of 0.0720, indicating moderate evidence that the regime-mean specification performs better than KF–MS–LSTM in this market. MS-LSTM, MS-GRU, and KF–MS–GRU also produce positive *t*-statistics, but their MBF values exceed 0.10, suggesting weak evidence and no decisive statistical difference. For Nikkei 225, the comparisons among regime-aware models are statistically much closer. Standard MS, MS-LSTM, MS-GRU, and KF–MS–GRU all generate MBF values above 0.10, indicating weak evidence of performance differences relative to KF–MS–LSTM. By contrast, KF–MS–Transformer records a *t*-statistic of −3.0201 and an MBF of 0.0720, suggesting moderate evidence in favor of KF–MS–LSTM over the attention-based regime-aware alternative.

Table 12 reports the economic value of the trading strategies for the S&P 500 and Nikkei 225 under a 0.1% transaction-cost assumption. Overall, the results show that regime-aware models consistently outperform non-regime benchmarks in both markets, although the leading specification differs across market environments.

Table 12. Economic Value of Trading Strategies for S&P 500 and Nikkei 225 with TC (0.1%).

Model A	Cumulative Return (%)	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio	MDD	Win Rate	Turnover	Calmar	Info Ratio
S&P 500									
ARIMA	−0.1162	−0.0379	0.1329	−0.2851	−0.2093	0.3201	0.6365	−0.1810	−2.7769
GARCH	0.7020	0.1809	0.1535	1.1789	−0.1958	0.5298	0.0658	0.9239	−0.9092
Standard MS	6.4450	0.8732	0.1067	8.1840	−0.0346	0.4107	0.1650	25.2327	4.2977
Pure LSTM	0.1265	0.0379	0.1239	0.3064	−0.1799	0.3226	0.3672	0.2109	−1.6330
GARCH-LSTM	0.1782	0.0526	0.1215	0.4331	−0.1561	0.1998	0.1898	0.3370	−1.4360
MS-LSTM	2.9662	0.5385	0.1194	4.5096	−0.0785	0.4107	0.2506	6.8580	2.4744
KF-MS-LSTM	2.7948	0.5174	0.1122	4.6130	−0.0810	0.3921	0.2457	6.3843	2.1407
Pure GRU	0.4860	0.1318	0.1507	0.8747	−0.2026	0.4826	0.1600	0.6507	−1.6219
MS-GRU	3.4540	0.5953	0.1162	5.1242	−0.0785	0.4057	0.3102	7.5817	2.7796
KF-MS-GRU	3.7538	0.6281	0.1151	5.4576	−0.0872	0.4032	0.2419	7.2048	2.9793
Transformer	0.8612	0.2144	0.1563	1.3715	−0.1890	0.5533	0.0000	1.1341	0.0000
KF-MS-Transformer	2.6050	0.4932	0.1261	3.9110	−0.1335	0.4094	0.2593	3.6952	2.3759
Buy & Hold	0.8612	0.2144	0.1563	1.3715	−0.1890	0.5533	1.0000	1.1341	0.0000
Nikkei 225									
ARIMA	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	−0.9073
GARCH	0.9582	0.2418	0.2185	1.1068	−0.2633	0.5409	0.0256	0.9184	0.4871
Standard MS	10.8104	1.2158	0.1459	8.3336	−0.0296	0.4143	0.1432	41.1159	3.9157
Pure LSTM	0.4641	0.1307	0.1850	0.7064	−0.2198	0.3261	0.2442	0.5946	−0.6377
GARCH-LSTM	−0.0692	−0.0228	0.1630	−0.1401	−0.2697	0.1100	0.0588	−0.0847	−1.4957
MS-LSTM	5.8464	0.8588	0.1629	5.2715	−0.1867	0.3875	0.3824	4.5997	2.9669
KF-MS-LSTM	9.8570	1.1565	0.1655	6.9862	−0.1960	0.4079	0.3593	5.9019	4.3280
Pure GRU	0.0489	0.0155	0.1797	0.0863	−0.2959	0.3056	0.5038	0.0524	−1.4263
MS-GRU	7.4646	0.9903	0.1623	6.1027	−0.2098	0.4028	0.4130	4.7201	3.4993
KF-MS-GRU	7.8087	1.0160	0.1555	6.5332	−0.1842	0.3875	0.3619	5.5148	3.4183
Transformer	0.8639	0.222	0.2212	1.0047	−0.2626	0.5486	0.0000	0.8463	0.0000
KF-MS-Transformer	3.7071	0.6474	0.1785	3.6275	−0.1847	0.3798	0.2647	3.5044	2.3746
Buy & Hold	0.8639	0.222	0.2212	1.0047	−0.2626	0.5486	1.0000	0.8463	0.0000

For the S&P 500, Standard MS achieves the strongest overall risk-adjusted performance, recording the highest cumulative return (644.50%), the highest annualized return (87.32%), the highest Sharpe ratio (8.1840), and the lowest maximum drawdown (−3.46%). KF–MS–GRU also performs strongly, with a cumulative return of 375.38%, an annualized return of 62.81%, and a Sharpe ratio of 5.4576. MS-GRU, MS-LSTM, KF–MS–LSTM, and KF–MS–

Transformer all substantially outperform the Buy-and-Hold benchmark, which records a cumulative return of only 86.12% and a Sharpe ratio of 1.3715. By contrast, ARIMA produces a negative cumulative return (−11.62%), while Pure LSTM, Pure GRU, and GARCH-LSTM generate only limited economic gains. These results indicate that regime conditioning contributes more to trading profitability than neural-network complexity alone in the U.S. market.

The S&P 500 results also suggest that relatively simple regime-based structures can capture substantial market-timing information. Standard MS achieves both the lowest drawdown and the highest Calmar ratio (25.2327), indicating strong downside-risk control. Although KF–MS–GRU and KF–MS–LSTM incorporate additional nonlinear learning layers, their incremental gains over Standard MS are comparatively moderate. This pattern is consistent with the earlier forecasting and DM-test results, where regime information itself provides the dominant source of predictive improvement in the U.S. market.

For the Nikkei 225, the economic-performance ranking differs somewhat from the S&P 500. KF–MS–LSTM achieves the highest cumulative return (985.70%) and the highest annualized return (115.65%), while KF–MS–GRU also performs strongly, with a cumulative return of 780.87%, an annualized return of 101.60%, and a Sharpe ratio of 6.5332. Standard MS records the highest Sharpe ratio (8.3336), the lowest maximum drawdown (−2.96%), and the highest Calmar ratio (41.1159), indicating the strongest downside-risk-adjusted performance. MS-LSTM and MS-GRU also substantially outperform Pure LSTM and Pure GRU, further confirming the importance of regime information in the Japanese market.

The Nikkei 225 results reveal a clearer trade-off between return generation and drawdown control. KF–MS–LSTM captures stronger upside potential but experiences a larger maximum drawdown (−19.60%), whereas Standard MS provides more conservative but highly stable risk-adjusted performance. Compared with the Buy-and-Hold benchmark, which records a cumulative return of 86.39% and a Sharpe ratio of 1.0047, all major regime-aware models generate substantially stronger economic value. In contrast, ARIMA again produces zero trading returns, while GARCH-LSTM delivers negative cumulative performance (−6.92%).

Taken together, the cross-market economic results reinforce the earlier statistical evidence from Tables 10 and 11. Across both international markets, regime-aware models consistently dominate non-regime benchmarks in cumulative return, Sharpe ratio, and downside-risk control. However, the best-performing specification remains market dependent: Standard MS dominates in risk-adjusted performance, particularly in the S&P 500, while KF–MS–LSTM and KF–MS–GRU provide stronger upside capture in the Nikkei 225. These findings suggest that regime conditioning is the primary source of economic value, whereas the incremental benefits of Kalman filtering and recurrent nonlinear learning depend on the underlying market structure.

3.7.4. Robustness Check: Transaction-Cost Sensitivity Across Markets

To provide a more realistic assessment of the cross-market economic results, this subsection further examines the sensitivity of trading performance to alternative transaction-cost assumptions. The detailed results for the S&P 500 and Nikkei 225 are reported in Tables A7 and A8 in Appendix C.

Overall, annualized returns and Sharpe ratios decline as transaction costs increase, particularly for models with relatively high turnover ratios. Nevertheless, the main comparative findings remain stable across both markets: regime-aware models continue to outperform most non-regime benchmarks under low and moderate transaction-cost assumptions. This suggests that the economic value of the proposed framework is not driven solely by frictionless trading assumptions.

For the S&P 500, Standard MS remains one of the most robust specifications because of its relatively low turnover ratio (0.1650) and strong downside-risk control. In the baseline setting reported in Table 12, it achieves the highest Sharpe ratio (8.1840) and the lowest maximum drawdown (−3.46%). KF–MS–GRU and KF–MS–LSTM also maintain strong economic performance, with cumulative returns of 375.38% and 279.48%, respectively, although their profitability becomes more sensitive as transaction costs increase because of their higher turnover ratios. By contrast, ARIMA, Pure LSTM, Pure GRU, and GARCH-LSTM remain economically weak even before imposing high transaction costs.

A similar pattern appears in the Nikkei 225 market. KF–MS–LSTM achieves the highest cumulative return (985.70%) and annualized return (115.65%) under the baseline cost assumption, while KF–MS–GRU records a cumulative return of 780.87% with a Sharpe ratio of 6.5332. However, these hybrid models also exhibit relatively high turnover ratios, causing profitability to deteriorate more rapidly under higher trading-cost assumptions. In contrast, Standard MS remains comparatively stable because of its lower turnover (0.1432), highest Sharpe ratio (8.3336), and smallest maximum drawdown (−2.96%).

These findings provide a more conservative interpretation of the economic-validation results. The high Sharpe ratios reported in Table 12 should not be interpreted as fully implementable institutional trading performance. Instead, they should be viewed as evidence of relative economic value under a simplified long-only index-timing framework. After accounting for transaction-cost sensitivity, the central conclusion remains unchanged: regime conditioning consistently improves the economic usefulness of forecasting models across markets, although the realized profitability of each specification depends on turnover, execution frictions, and market-specific trading conditions.

Taken together, the cross-market validation results confirm that the proposed regime-aware forecasting framework remains effective beyond the CSI 300 market, although the best-performing specification varies across market environments. Across both the S&P 500 and Nikkei 225, models incorporating Markov-switching regime information consistently outperform classical econometric, volatility-based, pure deep-learning, and standalone Transformer benchmarks in forecasting accuracy, DM-test evidence, and economic value. These findings suggest that regime conditioning is the primary source of robustness across heterogeneous financial markets, while the incremental contribution of Kalman filtering and nonlinear recurrent learning remains market dependent.

4. Discussion

The empirical results across three markets provide several insights into the role of regime-aware hybrid models in financial forecasting. An important finding of this study is that regime conditioning is the most consistent source of forecasting improvement and economic value across markets. The performance gap between regime-conditional models, Standard MS, KF-MS-LSTM, KF-MS-GRU, and non-regime-conditional models is large and statistically significant on every market and every evaluation criterion.

On CSI 300, the KF-MS-LSTM framework performs strongly in both statistical and economic terms. It achieves the lowest RMSE and MAE among the main forecasting models, while also generating a high Sharpe ratio and strong cumulative return in the out-of-sample trading simulation. The ablation results show that the largest improvement occurs when Markov-switching regime information is added to the recurrent models. Moving from Pure LSTM or Pure GRU to their MS-augmented versions produces a substantial improvement in forecasting accuracy and directional performance. This finding is consistent with Ang and Bekaert [40] and Guidolin and Timmermann [50], who demonstrate that regime-switching models capture structural dynamics that linear specifications miss, and extend these results to the context of deep sequential learning. The additional KF–MS structure

further improves the results, but its contribution is more incremental. Therefore, the CSI 300 evidence supports the view that the proposed hybrid framework is useful, while also showing that the dominant mechanism is the extraction of regime-dependent information rather than the LSTM architecture alone.

The cross-market evidence further shows that model performance varies across financial systems. In the S&P 500, Standard MS performs particularly strongly, achieving the best forecasting accuracy and the highest Sharpe ratio among the models. This suggests that, in a highly developed and relatively information-efficient market, a parsimonious regime signal can already capture a large portion of the exploitable predictability. In contrast, for the Nikkei 225, KF–MS–LSTM and KF–MS–GRU become more competitive. KF–MS–LSTM delivers the strongest meaningful directional accuracy, while Standard MS retains the best drawdown control and Sharpe ratio. Thus, the cross-market validation results support the robustness of the regime-aware framework, but not the unconditional dominance of a single architecture. The relative value of LSTM, GRU, Transformer, and Standard MS components depends on the market structure, volatility patterns, and strength of regime-dependent signals.

The Kalman filter's role deserves careful qualification. The results suggest that the Kalman Filter mainly serves as a refinement tool within the proposed framework. By smoothing short-term noise and generating filtered returns, it provides cleaner inputs for Markov Switching regime identification and helps improve the quality of regime-based signals [55]. However, its contribution is market-dependent. The KF–MS specifications improve performance in some cases, such as the CSI 300 and Nikkei 225, but Standard MS remains highly competitive in the S&P 500. Therefore, the Kalman Filter should be interpreted as a useful preprocessing component under noisy or volatile market conditions, rather than as a mechanism that uniformly improves performance across all markets.

The comparison between LSTM and GRU models provides a useful practical implication. Across the CSI 300 and the two validation markets, GRU-based variants often match or exceed LSTM-based variants in selected forecasting and trading metrics, suggesting that the choice of recurrent cell is less important than the inclusion of regime information. This finding is consistent with recent comparative studies showing that LSTM and GRU often produce similar forecasting performance, while GRU can be computationally attractive because of its simpler gating structure and lower parameter complexity. Recent stock-forecasting studies also show that GRU may outperform LSTM in some datasets, whereas LSTM may perform better in others, implying that the relative advantage of the two architectures is data- and market-dependent rather than universal [56,57].

Although the high Sharpe ratios and cumulative returns indicate strong relative economic value under the long trading rule, they should not be viewed as directly achievable institutional trading performance. The transaction-cost sensitivity analysis shows that profitability declines as trading frictions increase, particularly for higher-turnover neural hybrid models, while the tail-exclusion results indicate that performance is not driven solely by a few extreme observations. The additional Sharpe stability diagnostics further support this interpretation: the main regime-aware models retain positive and high Sharpe ratios under block-bootstrap resampling, leave-one-out exclusion, and top 1% return-day removal, suggesting that the reported risk-adjusted performance is not explained by a single extreme trading day. However, the simulations do not fully account for slippage, market impact, liquidity constraints, taxes, short-sale restrictions, or execution delays. Therefore, the main economic conclusion is comparative: under the same trading rule and cost assumptions, regime-aware models provide more useful trading signals than non-regime benchmarks.

This study's findings contribute to the financial forecasting literature in three ways. First, they show that regime-switching information remains highly valuable when combined with modern machine-learning models. Second, they demonstrate that more complex deep-learning architectures do not necessarily outperform simpler regime-based models unless they are supported by meaningful market-state information. Third, they show that statistical forecasting accuracy and economic performance are related but not identical: the model with the lowest forecast error is not always the model with the best drawdown control or Sharpe ratio. The study therefore supports a more nuanced conclusion: regime-aware forecasting improves robustness across markets, but the optimal specification depends on the evaluation objective, such as magnitude accuracy, directional prediction, return generation, or risk control [35,36].

5. Conclusions

This study proposes a hybrid regime-aware forecasting framework that integrates Kalman Filtering, Markov Switching, and recurrent neural networks for stock-index forecasting. The Kalman Filter is used to reduce short-term observational noise, the Markov Switching model identifies latent market regimes, and the recurrent neural network components capture nonlinear sequential patterns in regime-conditioned information. The model is evaluated on three major equity indices across three distinct institutional environments: the CSI 300 (China), S&P 500 (United States), and Nikkei 225 (Japan), using both statistical accuracy metrics and out-of-sample trading simulations.

The main finding is that regime conditioning is the most stable source of forecasting improvement and economic value across all three markets. On the CSI 300, KF–MS–LSTM achieves the strongest overall forecasting accuracy, with the lowest RMSE and MAE among the benchmark models, while the DM test shows decisive evidence that it outperforms classical, volatility-based, pure deep-learning, and standalone Transformer benchmarks. The economic results also show that regime-aware models generate substantially stronger cumulative returns and Sharpe ratios than non-regime models. However, the results also indicate that the improvement from Kalman filtering and recurrent learning is incremental once regime information has already been introduced.

The cross-market validation further confirms the robustness of the regime-aware approach, while also showing that the best-performing specification is market-dependent. In S&P 500, Standard MS performs particularly strongly in both forecasting accuracy and economic value. However, In the Nikkei 225, KF–MS–LSTM and KF–MS–GRU become more competitive. These findings show that the relative value of LSTM, GRU, Transformer, and Standard MS components depends on market-specific dynamics, including volatility structure, regime persistence, and the complexity of short-term return behavior.

From the results, it also offers important implications for model design. Pure deep-learning models, such as LSTM, GRU, and Transformer, perform weakly when used without regime information. In contrast, models using Markov-switching probabilities or regime-mean signals achieve stronger forecasting and trading performance across markets. This suggests that financial forecasting benefits from combining nonlinear learning with interpretable regime information. The results also show that GRU is a competitive alternative to LSTM, as KF–MS–GRU often matches or exceeds KF–MS–LSTM while using a simpler recurrent structure.

For a practical perspective, the regime-aware framework provides useful investment signals under the same trading rules. However, the high Sharpe ratios and cumulative returns should be interpreted cautiously because the simulations use a simplified long strategy and do not fully include slippage, taxes, liquidity constraints, market impact, financing costs, or execution delays. Since profitability declines when transaction costs

increase, the results should be viewed as comparative evidence that regime-conditioned forecasts contain economic value, rather than as directly achievable trading performance. At the same time, the added Sharpe stability diagnostics show that the main regime-aware models remain robust after bootstrap resampling, leave-one-out testing, and excluding the top 1% of strategy-return days. This strengthens the conclusion that the economic value is not driven solely by isolated extreme observations, although large directional market movements still contribute to profitability.

For policymakers and market regulators, the regime-identification component provides a potentially useful indicator for monitoring changes in market conditions. The three-state classification of adverse return-volatility, intermediate market, and favorable return-volatility states offers an interpretable way to summarize shifts in market risk and may complement traditional indicators such as volatility, drawdowns, and liquidity measures. However, the results should be interpreted as evidence from an empirical forecasting framework rather than as a fully developed regulatory early-warning system. Future research could further examine whether regime probabilities can be integrated with market-stability indicators to support real-time risk monitoring and policy assessment.

However, this study has several limitations that suggest directions for future research. First, the trading strategy uses a simple long-only rule with fixed transaction-cost assumptions. Future research could examine regime-specific position sizing, volatility-adjusted exposure, and probability-based trading thresholds to reduce turnover during uncertain transition periods. Second, the same three-state Markov Switching structure is applied across all markets for comparability. Future studies could allow the number of regimes or transition probabilities to vary across markets, possibly through Bayesian regime-switching or time-varying transition models. Third, the current feature set is based mainly on price-derived variables, including raw returns, Kalman-filtered returns, regime probabilities, and regime-based signals. Incorporating macroeconomic indicators, investor sentiment, option-implied information, or cross-asset variables may improve forecasting performance during periods of macro-financial stress. Future research may further extend the framework to rolling-window estimation, multi-step forecasting, and crisis-specific subsample evaluation.

This study demonstrates that regime-aware forecasting provides a robust and interpretable framework for stock prediction across heterogeneous markets. The empirical evidence consistently shows that incorporating latent market-state information improves both forecasting accuracy and economic performance relative to conventional linear, volatility-based, and standalone deep-learning benchmarks. However, the optimal specification varies across market environments and evaluation objectives, including magnitude accuracy, directional prediction, return generation, and downside-risk control. In particular, the results indicate that regime conditioning is the most stable source of forecasting improvement, while the incremental value of Kalman filtering and specific neural-network architectures remains market-dependent.

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Appendix A

Table A1. Regime identification results.

Market	K = 2 (AIC)	K = 3 (AIC)	K = 2 (BIC)	K = 3 (BIC)	K = 2 (LL)	K = 3 (LL)
CSI 300	−18,460.07	**−18,829.22**	−18,411.74	**−18,738.61**	9238.03	**9429.61**
S&P 500	−21,256.75	**−21,860.73**	−21,208.13	**−21,769.57**	10,636.37	**10,945.36**
Nikkei 225	−18,892.96	**−19,351.27**	−18,844.59	**−19,260.56**	9454.48	**9690.63**

Note: Bold figure means the best results for each criterion.

Table A2. Hyperparameter Configuration—All Neural Network Models.

Hyperparameter	Value	Notes
Lookback window (L)	5 trading days	One calendar week; robust to L ∈ {5, 10, 20}
LSTM architecture	128 → 64 → 32 → Dense (1)	Three stacked LSTM layers + output
GRU architecture	128 → 64 → 32 → Dense (1)	Identical structure to the LSTM variant
Dropout rates	0.20/0.20/0.10	After each recurrent layer, respectively
Residual skip	MS signal added to Dense output	Guarantees model ≥ Standard MS accuracy
Optimizer	Adam	Default learning rate 0.001
Batch size	32	
Max epochs	150	With EarlyStopping (patience = 15)
LR schedule	ReduceLROnPlateau	Factor = 0.5, patience = 10
Loss function	Mean Squared Error	On log-returns
Train/test split	80%/20%	Chronological; no shuffle

Note: All models (Pure LSTM, MS-LSTM, KF-MS-LSTM, and GRU variants) use the same architecture and training configuration to ensure a fair ablation. ARIMA orders are selected by grid search over p ∈ {0,1,2,3,4} and q ∈ {0, 1, 2, 3, 4}, minimizing AIC on the training set.

Appendix B

Table A3. Transaction-Cost Sensitivity Analysis for CSI 300.

Model	TC	Annual Return	Sharpe	MDD	Turnover
ARIMA	0.05%	0.0000	0.0000	0.0000	0.0000
ARIMA	0.20%	0.0000	0.0000	0.0000	0.0000
ARIMA	0.50%	0.0000	0.0000	0.0000	0.0000
ARIMA	1.00%	0.0000	0.0000	0.0000	0.0000
GARCH	0.05%	−0.0004	−0.0026	−0.3149	0.3012
GARCH	0.20%	−0.1079	−0.7128	−0.4354	0.3012
GARCH	0.50%	−0.2896	−1.8689	−0.6664	0.3012
GARCH	1.00%	−0.5139	−3.0752	−0.8904	0.3012
Standard MS	0.05%	0.7225	5.5744	−0.0705	0.1236
Standard MS	0.20%	0.6439	5.0002	−0.0705	0.1236
Standard MS	0.50%	0.4973	3.8690	−0.0705	0.1236
Standard MS	1.00%	0.2814	2.1269	−0.0705	0.1236
Pure LSTM	0.05%	−0.0255	−0.1862	−0.2873	0.3423
Pure LSTM	0.20%	−0.1438	−1.0400	−0.4321	0.3423
Pure LSTM	0.50%	−0.3390	−2.3636	−0.7340	0.3423
Pure LSTM	1.00%	−0.5706	−3.5893	−0.9271	0.3423
GARCH-LSTM	0.05%	0.0558	0.3406	−0.2480	0.0103
GARCH-LSTM	0.20%	0.0517	0.3155	−0.2480	0.0103

Table A3. *Cont.*

Model	TC	Annual Return	Sharpe	MDD	Turnover
GARCH-LSTM	0.50%	0.0436	0.2654	−0.2480	0.0103
GARCH-LSTM	1.00%	0.0301	0.1827	−0.2480	0.0103
MS-LSTM	0.05%	0.8237	6.3655	−0.0945	0.3166
MS-LSTM	0.20%	0.6180	4.7310	−0.0958	0.3166
MS-LSTM	0.50%	0.2736	2.0148	−0.1040	0.3166
MS-LSTM	1.00%	−0.1454	−0.9604	−0.3974	0.3166
KF-MS-LSTM	0.05%	0.8491	6.5477	−0.0867	0.2857
KF-MS-LSTM	0.20%	0.6598	5.0367	−0.0867	0.2857
KF-MS-LSTM	0.50%	0.3374	2.4774	−0.0916	0.2857
KF-MS-LSTM	1.00%	−0.0669	−0.4422	−0.2798	0.2857
Pure GRU	0.05%	0.0180	0.1341	−0.2619	0.3719
Pure GRU	0.20%	−0.1155	−0.8547	−0.4113	0.3719
Pure GRU	0.50%	−0.3323	−2.3679	−0.7247	0.3719
Pure GRU	1.00%	−0.5821	−3.7244	−0.9346	0.3719
MS-GRU	0.05%	0.8103	6.1927	−0.1173	0.3037
MS-GRU	0.20%	0.6139	4.6558	−0.1200	0.3037
MS-GRU	0.50%	0.2828	2.0712	−0.1281	0.3037
MS-GRU	1.00%	−0.1251	−0.8272	−0.3646	0.3037
KF-MS-GRU	0.05%	0.9162	7.6154	−0.0277	0.3269
KF-MS-GRU	0.20%	0.6935	5.7100	−0.0277	0.3269
KF-MS-GRU	0.50%	0.3227	2.5468	−0.0534	0.3269
KF-MS-GRU	1.00%	−0.1239	−0.8665	−0.3885	0.3269
Transformer	0.05%	0.0714	0.4138	−0.2480	0.0000
Transformer	0.20%	0.0714	0.4138	−0.2480	0.0000
Transformer	0.50%	0.0714	0.4138	−0.2480	0.0000
Transformer	1.00%	0.0714	0.4138	−0.2480	0.0000
KF-MS-Transformer	0.05%	0.5404	4.2201	−0.1082	0.2008
KF-MS-Transformer	0.20%	0.4278	3.3323	−0.1136	0.2008
KF-MS-Transformer	0.50%	0.2267	1.7288	−0.1242	0.2008
KF-MS-Transformer	1.00%	−0.0475	−0.3355	−0.2135	0.2008

Note: TC denotes proportional transaction cost per trade. Annual Return, Sharpe, MDD, and Turnover denote annualized return, Sharpe ratio, maximum drawdown, and average turnover, respectively.

Table A4. Tail-Exclusion Diagnostics for CSI 300.

Model	Tail Excluded	Remaining Observations	Annual Return	Annual Volatility	Sharpe
ARIMA	1% each tail	761	0.0000	0.0000	0.0000
ARIMA	5% each tail	699	0.0000	0.0000	0.0000
ARIMA	10% each tail	621	0.0000	0.0000	0.0000
GARCH	1% each tail	761	−0.0687	0.1167	−0.5883
GARCH	5% each tail	699	−0.0903	0.0873	−1.0342
GARCH	10% each tail	621	−0.1186	0.0678	−1.7503
Standard MS	1% each tail	761	0.6050	0.0940	6.4350
Standard MS	5% each tail	699	0.4418	0.0638	6.9230
Standard MS	10% each tail	621	0.3171	0.0437	7.2605
Pure LSTM	1% each tail	761	−0.0862	0.0894	−0.9645
Pure LSTM	5% each tail	699	−0.1134	0.0565	−2.0083
Pure LSTM	10% each tail	621	−0.1320	0.0337	−3.9158
GARCH-LSTM	1% each tail	761	0.0346	0.1274	0.2715
GARCH-LSTM	5% each tail	699	0.0119	0.0984	0.1206
GARCH-LSTM	10% each tail	621	−0.0110	0.0784	−0.1400
MS-LSTM	1% each tail	761	0.6801	0.0932	7.2982
MS-LSTM	5% each tail	699	0.5391	0.0677	7.9679

Table A4. *Cont.*

Model	Tail Excluded	Remaining Observations	Annual Return	Annual Volatility	Sharpe
MS-LSTM	10% each tail	621	0.4077	0.0497	8.2105
KF-MS-LSTM	1% each tail	761	0.6952	0.0944	7.3609
KF-MS-LSTM	5% each tail	699	0.5430	0.0668	8.1290
KF-MS-LSTM	10% each tail	621	0.4097	0.0490	8.3649
Pure GRU	1% each tail	761	−0.0656	0.0949	−0.6914
Pure GRU	5% each tail	699	−0.0851	0.0657	−1.2947
Pure GRU	10% each tail	621	−0.1144	0.0449	−2.5476
MS-GRU	1% each tail	761	0.6699	0.0950	7.0553
MS-GRU	5% each tail	699	0.5265	0.0683	7.7135
MS-GRU	10% each tail	621	0.4013	0.0498	8.0583
KF-MS-GRU	1% each tail	761	0.7127	0.0915	7.7902
KF-MS-GRU	5% each tail	699	0.5402	0.0638	8.4681
KF-MS-GRU	10% each tail	633	0.3479	0.0465	7.4761
Transformer	1% each tail	761	0.0538	0.1376	0.3909
Transformer	5% each tail	699	0.0191	0.1064	0.1797
Transformer	10% each tail	621	−0.0064	0.0865	−0.0745
KF-MS-Transformer	1% each tail	761	0.4295	0.0905	4.7444
KF-MS-Transformer	5% each tail	699	0.3093	0.0620	4.9913
KF-MS-Transformer	10% each tail	621	0.2075	0.0408	5.0910

Table A5. Lookback Sensitivity Analysis for CSI 300.

Lookback	RMSE	MAE	TIC	MDA	R^2	Cumulative Return	Sharpe	MDD	Turnover
5	34.7702	21.8597	0.0045	0.7362	0.9903	4.9559	6.0276	−0.0867	0.2857
10	35.6523	22.7610	0.0046	0.7001	0.9898	3.7770	5.0940	−0.0949	0.2175
20	36.1439	23.8904	0.0047	0.7104	0.9895	4.2674	5.6641	−0.0867	0.2394

Table A6. Sharpe Stability Diagnostics for CSI 300 Trading Strategies (Observations = 777).

Model	Baseline Sharpe	Bootstrap Mean Sharpe	Bootstrap SD	95% CI Low	95% CI High	LOO Min Sharpe	LOO Max Sharpe	Sharpe No Top 1%	Sharpe No Bottom 1%	Sharpe No Both 1%
ARIMA	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
GARCH	−0.2491	−0.1868	0.6723	−1.3521	1.1868	−0.4368	−0.1001	−1.1193	0.3527	−0.5883
Standard MS	5.3828	5.5353	0.8008	4.0259	7.1048	5.2932	6.0460	4.8420	6.6775	6.4350
Pure LSTM	−0.4853	−0.4964	0.6765	−1.8213	0.8496	−0.7046	−0.3365	−1.5301	0.2655	−0.9645
GARCH-LSTM	0.3322	0.4172	0.6635	−0.7719	1.7359	0.1717	0.5036	−0.4476	1.1210	0.2715
MS-LSTM	5.8007	5.9646	0.9378	4.2507	7.7236	5.7116	6.4983	5.2917	7.4214	7.2982
KF-MS-LSTM	6.0276	6.1896	0.8994	4.4557	7.9912	5.9386	6.7431	5.5317	7.5035	7.3609
Pure GRU	−0.2132	−0.2133	0.7141	−1.5101	1.2437	−0.4285	−0.0417	−1.2564	0.4354	−0.6914
MS-GRU	5.6629	5.8238	0.9221	4.1533	7.6081	5.5728	6.3381	5.1259	7.2421	7.0553
KF-MS-GRU	6.9573	7.1064	0.6958	5.8140	8.5618	6.8744	7.1794	6.7348	7.8162	7.7902
Transformer	0.4138	0.5049	0.6673	−0.6799	1.9313	0.2618	0.5806	−0.3281	1.2016	0.3909
KF-MS-Transformer	3.9186	4.0252	0.7868	2.5626	5.6236	3.8248	4.4581	3.2539	5.1466	4.7444

Note: Bootstrap Sharpe statistics are based on 1000 block-bootstrap replications with a five-day block length and 0.1% transaction cost. "LOO" denotes leave-one-out diagnostics, where the Sharpe ratio is recalculated after removing each trading day one at a time. "No Top 1%," "No Bottom 1%," and "No Both 1%" report Sharpe ratios after excluding the top 1%, bottom 1%, and both tails of daily strategy returns, respectively.

Appendix C

Table A7. Transaction-Cost Sensitivity Analysis for S&P 500.

Model	TC	Annual Return	Sharpe	MDD	Turnover
ARIMA	0.50%	−0.4935	−3.5392	−0.8867	0.6365
ARIMA	1.00%	−0.7729	−4.9596	−0.9913	0.6365
GARCH	0.50%	0.1052	0.6801	−0.2022	0.0658
GARCH	1.00%	0.0173	0.1092	−0.2101	0.0658
Standard MS	0.50%	0.5862	5.5798	−0.0346	0.1650
Standard MS	1.00%	0.2884	2.6149	−0.0363	0.1650
Pure LSTM	0.50%	−0.2832	−2.2013	−0.6717	0.3672
Pure LSTM	1.00%	−0.5487	−3.8236	−0.9236	0.3672
GARCH-LSTM	0.50%	−0.1307	−1.0446	−0.4327	0.1898
GARCH-LSTM	1.00%	−0.3156	−2.3184	−0.7071	0.1898
MS-LSTM	0.50%	0.1950	1.5512	−0.0997	0.2506
MS-LSTM	1.00%	−0.1286	−0.9122	−0.4125	0.2506
KF-MS-LSTM	0.50%	0.1845	1.5690	−0.1025	0.2457
KF-MS-LSTM	1.00%	−0.1308	−0.9887	−0.3892	0.2457
Pure GRU	0.50%	−0.0368	−0.2393	−0.2734	0.3102
Pure GRU	1.00%	−0.2127	−1.3107	−0.5459	0.1600
MS-GRU	0.50%	0.1669	1.3432	−0.0859	0.2419
MS-GRU	1.00%	−0.2106	−0.1477	−0.5379	0.3102
KF-MS-GRU	0.50%	0.2758	2.3078	−0.1017	0.2419
KF-MS-GRU	1.00%	−0.0595	−0.4475	−0.2306	0.2419
Transformer	0.50%	0.2144	1.3715	−0.1890	0.0000
Transformer	1.00%	0.2144	1.3715	−0.1890	0.0000
KF-MS-Transformer	0.50%	0.1497	1.1450	−0.1785	0.2593
KF-MS-Transformer	1.00%	−0.1707	−1.1840	−0.4906	0.2593

Table A8. Transaction-Cost Sensitivity Analysis for Nikkei 225.

Model	TC	Annual Return	Sharpe	MDD	Turnover
ARIMA	0.50%	0.0000	0.0000	0.0000	0.0000
ARIMA	1.00%	0.0000	0.0000	0.0000	0.0000
GARCH	0.50%	0.2102	0.9626	−0.2865	0.0256
GARCH	1.00%	0.1718	0.7850	−0.3145	0.0256
Standard MS	0.50%	0.9180	6.3016	−0.0296	0.1432
Standard MS	1.00%	0.6013	4.0058	−0.0367	0.1432
Pure LSTM	0.50%	−0.1160	−0.6142	−0.4230	0.2442
Pure LSTM	1.00%	−0.3502	−1.7587	−0.7384	0.2442
GARCH-LSTM	0.50%	−0.0791	−0.4817	−0.3360	0.0588
GARCH-LSTM	1.00%	−0.1449	−0.8646	−0.4285	0.0588
MS-LSTM	0.50%	0.2643	1.5652	−0.2406	0.3824
MS-LSTM	1.00%	−0.2191	−1.1941	−0.6174	0.3824
KF-MS-LSTM	0.50%	0.5012	2.9544	−0.2213	0.3593
KF-MS-LSTM	1.00%	−0.0452	−0.2495	−0.3799	0.3593
Pure GRU	0.50%	−0.3889	−2.1027	−0.7845	0.5038
Pure GRU	1.00%	−0.6761	−3.4050	−0.9696	0.5038
MS-GRU	0.50%	0.3125	1.8538	−0.2285	0.4130
MS-GRU	1.00%	−0.2200	−1.1971	−0.5744	0.4130
KF-MS-GRU	0.50%	0.3998	2.4824	−0.2064	0.3619
KF-MS-GRU	1.00%	−0.1128	−0.6430	−0.4954	0.3619
Transformer	0.50%	0.2222	1.0047	−0.2626	0.0000
Transformer	1.00%	0.2222	1.0047	−0.2626	0.0000
KF-MS-Transformer	0.50%	0.2616	1.4182	−0.2041	0.2647
KF-MS-Transformer	1.00%	−0.0962	−0.4876	−0.3814	0.2647

Appendix D

Table A9. Repeated-Initialization Robustness of Neural-Network Forecasting Models.

Model	RMSE Mean	RMSE SD	MAE Mean	MAE SD	TIC Mean	TIC SD	MDA Mean	MDA SD	R^2 Mean	R^2 SD
CSI 300										
Pure LSTM	42.3359	0.2796	28.7124	0.0775	0.0055	0.0000	0.4819	0.0163	0.9856	0.0002
GARCH-LSTM	42.4224	0.2554	29.2703	0.3674	0.0055	0.0000	0.4965	0.0047	0.9855	0.0002
MS-LSTM	35.8257	1.1639	22.5640	0.5446	0.0046	0.0001	0.7201	0.0075	0.9897	0.0007
KF-MS-LSTM	35.3633	0.3854	22.3325	0.2538	0.0046	0.0001	0.7270	0.0116	0.9900	0.0002
Pure GRU	42.3213	0.1888	28.7762	0.0792	0.0055	0.0000	0.4865	0.0083	0.9856	0.0001
MS-GRU	35.6309	0.8222	22.5235	0.5789	0.0046	0.0001	0.7305	0.0116	0.9898	0.0005
KF-MS-GRU	35.2826	0.3015	22.0783	0.3359	0.0046	0.0000	0.7281	0.0061	0.9900	0.0002
Transformer	42.0665	0.0006	28.6321	0.0002	0.0054	0.0000	0.4968	0.0000	0.9858	0.0000
KF-MS-Transformer	38.3227	0.6204	25.9357	0.2598	0.0050	0.0001	0.6377	0.0092	0.9882	0.0004
S&P 500										
Pure LSTM	49.8693	0.3812	34.9177	0.0497	0.0047	0.0000	0.5189	0.0138	0.9970	0.0000
GARCH-LSTM	50.3573	0.1840	35.2778	0.3672	0.0048	0.0000	0.4887	0.0148	0.9970	0.0000
MS-LSTM	43.8857	1.1357	29.5661	1.0077	0.0041	0.0001	0.6682	0.0168	0.9977	0.0001
KF-MS-LSTM	45.4750	0.3489	31.1413	0.2404	0.0043	0.0000	0.6632	0.0163	0.9975	0.0000
Pure GRU	50.3003	0.1331	35.1270	0.1537	0.0047	0.0000	0.5104	0.0134	0.9970	0.0000
MS-GRU	45.0968	0.7253	30.2206	1.0023	0.0043	0.0001	0.6666	0.0210	0.9976	0.0001
KF-MS-GRU	44.8751	0.5379	30.6279	0.4227	0.0042	0.0001	0.6669	0.0158	0.9976	0.0001
Transformer	50.4026	0.0032	34.8471	0.0053	0.0048	0.0000	0.5558	0.0000	0.9970	0.0000
KF-MS-Transformer	46.8327	0.8346	31.8691	0.8616	0.0044	0.0001	0.6437	0.0179	0.9974	0.0001
Nikkei 225										
Pure LSTM	509.6773	0.7181	342.1455	1.0025	0.0069	0.0000	0.5387	0.0149	0.9929	0.0000
GARCH-LSTM	513.1683	2.7955	346.5812	3.6477	0.0070	0.0000	0.4714	0.0151	0.9928	0.0001
MS-LSTM	445.4176	7.5518	271.9171	3.2677	0.0060	0.0001	0.7261	0.0102	0.9946	0.0002
KF-MS-LSTM	435.2948	4.7918	264.5060	2.4236	0.0059	0.0001	0.7375	0.0077	0.9948	0.0001
Pure GRU	510.2184	1.1300	343.5722	1.4096	0.0069	0.0000	0.5096	0.0194	0.9929	0.0000
MS-GRU	439.0109	6.3117	269.0609	3.7909	0.0060	0.0001	0.7375	0.0111	0.9947	0.0002
KF-MS-GRU	427.9205	2.0777	261.3795	1.5628	0.0058	0.0000	0.7444	0.0072	0.9950	0.0000
Transformer	509.9384	0.0774	341.8223	0.1485	0.0069	0.0000	0.5486	0.0000	0.9929	0.0000
KF-MS-Transformer	474.8882	5.2594	307.4796	4.7967	0.0064	0.0001	0.6656	0.0131	0.9938	0.0001

Note: Each stochastic neural-network model is estimated 10 times using different random initial weights. The chronological train–test split, input variables, model architecture, optimizer, early-stopping rule, and test dataset are held fixed across runs. The table reports the mean and standard deviation of RMSE, MAE, TIC, MDA, and R^2 on the test dataset. Lower RMSE, MAE, and TIC indicate better magnitude accuracy, while higher MDA and R^2 indicate stronger directional and explanatory performance.

Appendix E

Table A10. Empirical Distribution of Regime Spell Lengths for the CSI 300 Index.

Market	Regime	Regime Label	Spell Length (Days)	Frequency	Percentage (%)
CSI 300	0	Adverse return-volatility state	1	38	15.70
CSI 300	0	Adverse return-volatility state	2	43	17.77
CSI 300	0	Adverse return-volatility state	3	47	19.42
CSI 300	0	Adverse return-volatility state	4	29	11.98
CSI 300	0	Adverse return-volatility state	5	24	9.92
CSI 300	0	Adverse return-volatility state	6	18	7.44
CSI 300	0	Adverse return-volatility state	7	11	4.55
CSI 300	0	Adverse return-volatility state	8	11	4.55
CSI 300	0	Adverse return-volatility state	9	3	1.24
CSI 300	0	Adverse return-volatility state	10	7	2.89
CSI 300	0	Adverse return-volatility state	11	3	1.24
CSI 300	0	Adverse return-volatility state	12	2	0.83
CSI 300	0	Adverse return-volatility state	13	1	0.41
CSI 300	0	Adverse return-volatility state	15	2	0.83

Table A10. *Cont.*

Market	Regime	Regime Label	Spell Length (Days)	Frequency	Percentage (%)
CSI 300	0	Adverse return-volatility state	16	1	0.41
CSI 300	0	Adverse return-volatility state	19	2	0.83
CSI 300	1	Intermediate market state	1	101	22.39
CSI 300	1	Intermediate market state	2	109	24.17
CSI 300	1	Intermediate market state	3	81	17.96
CSI 300	1	Intermediate market state	4	64	14.19
CSI 300	1	Intermediate market state	5	44	9.76
CSI 300	1	Intermediate market state	6	16	3.55
CSI 300	1	Intermediate market state	7	8	1.77
CSI 300	1	Intermediate market state	8	6	1.33
CSI 300	1	Intermediate market state	9	8	1.77
CSI 300	1	Intermediate market state	10	6	1.33
CSI 300	1	Intermediate market state	11	2	0.44
CSI 300	1	Intermediate market state	12	1	0.22
CSI 300	1	Intermediate market state	13	1	0.22
CSI 300	1	Intermediate market state	14	1	0.22
CSI 300	1	Intermediate market state	15	1	0.22
CSI 300	1	Intermediate market state	16	1	0.22
CSI 300	1	Intermediate market state	20	1	0.22
CSI 300	2	Favorable return-volatility state	1	32	11.27
CSI 300	2	Favorable return-volatility state	2	34	11.97
CSI 300	2	Favorable return-volatility state	3	36	12.68
CSI 300	2	Favorable return-volatility state	4	47	16.55
CSI 300	2	Favorable return-volatility state	5	51	17.96
CSI 300	2	Favorable return-volatility state	6	26	9.15
CSI 300	2	Favorable return-volatility state	7	16	5.63
CSI 300	2	Favorable return-volatility state	8	11	3.87
CSI 300	2	Favorable return-volatility state	9	7	2.46
CSI 300	2	Favorable return-volatility state	10	6	2.11
CSI 300	2	Favorable return-volatility state	11	4	1.41
CSI 300	2	Favorable return-volatility state	12	4	1.41
CSI 300	2	Favorable return-volatility state	13	3	1.06
CSI 300	2	Favorable return-volatility state	14	3	1.06
CSI 300	2	Favorable return-volatility state	17	2	0.70
CSI 300	2	Favorable return-volatility state	20	1	0.35
CSI 300	2	Favorable return-volatility state	21	1	0.35

Table A11. Empirical Distribution of Regime Spell Lengths for the S&P 500 Index.

Market	Regime	Regime Label	Spell Length (Days)	Frequency	Percentage (%)
Nikkei 225	0	Adverse return-volatility state	1	43	16.80
Nikkei 225	0	Adverse return-volatility state	2	35	13.67
Nikkei 225	0	Adverse return-volatility state	3	51	19.92
Nikkei 225	0	Adverse return-volatility state	4	33	12.89
Nikkei 225	0	Adverse return-volatility state	5	33	12.89
Nikkei 225	0	Adverse return-volatility state	6	14	5.47
Nikkei 225	0	Adverse return-volatility state	7	21	8.20
Nikkei 225	0	Adverse return-volatility state	8	8	3.13
Nikkei 225	0	Adverse return-volatility state	9	3	1.17
Nikkei 225	0	Adverse return-volatility state	10	6	2.34
Nikkei 225	0	Adverse return-volatility state	11	3	1.17
Nikkei 225	0	Adverse return-volatility state	12	4	1.56
Nikkei 225	0	Adverse return-volatility state	14	1	0.39
Nikkei 225	0	Adverse return-volatility state	17	1	0.39

Table A11. *Cont.*

Market	Regime	Regime Label	Spell Length (Days)	Frequency	Percentage (%)
Nikkei 225	1	Intermediate market state	1	93	19.87
Nikkei 225	1	Intermediate market state	2	88	18.80
Nikkei 225	1	Intermediate market state	3	85	18.16
Nikkei 225	1	Intermediate market state	4	71	15.17
Nikkei 225	1	Intermediate market state	5	26	5.56
Nikkei 225	1	Intermediate market state	6	28	5.98
Nikkei 225	1	Intermediate market state	7	18	3.85
Nikkei 225	1	Intermediate market state	8	15	3.21
Nikkei 225	1	Intermediate market state	9	10	2.14
Nikkei 225	1	Intermediate market state	10	8	1.71
Nikkei 225	1	Intermediate market state	11	7	1.50
Nikkei 225	1	Intermediate market state	12	2	0.43
Nikkei 225	1	Intermediate market state	13	5	1.07
Nikkei 225	1	Intermediate market state	14	2	0.43
Nikkei 225	1	Intermediate market state	15	1	0.21
Nikkei 225	1	Intermediate market state	17	2	0.43
Nikkei 225	1	Intermediate market state	18	1	0.21
Nikkei 225	1	Intermediate market state	19	3	0.64
Nikkei 225	1	Intermediate market state	24	1	0.21
Nikkei 225	1	Intermediate market state	32	1	0.21
Nikkei 225	1	Intermediate market state	34	1	0.21
Nikkei 225	2	Favorable return-volatility state	1	37	15.88
Nikkei 225	2	Favorable return-volatility state	2	49	21.03
Nikkei 225	2	Favorable return-volatility state	3	38	16.31
Nikkei 225	2	Favorable return-volatility state	4	35	15.02
Nikkei 225	2	Favorable return-volatility state	5	19	8.15
Nikkei 225	2	Favorable return-volatility state	6	19	8.15
Nikkei 225	2	Favorable return-volatility state	7	16	6.87
Nikkei 225	2	Favorable return-volatility state	8	7	3.00
Nikkei 225	2	Favorable return-volatility state	9	3	1.29
Nikkei 225	2	Favorable return-volatility state	10	3	1.29
Nikkei 225	2	Favorable return-volatility state	11	1	0.43
Nikkei 225	2	Favorable return-volatility state	12	3	1.29
Nikkei 225	2	Favorable return-volatility state	13	1	0.43
Nikkei 225	2	Favorable return-volatility state	14	1	0.43
Nikkei 225	2	Favorable return-volatility state	25	1	0.43

Table A12. Empirical Distribution of Regime Spell Lengths for the Nikkei 225 Index.

Market	Regime	Regime Label	Spell Length (Days)	Frequency	Percentage (%)
S&P 500	0	Adverse return-volatility state	1	65	27.90
S&P 500	0	Adverse return-volatility state	2	52	22.32
S&P 500	0	Adverse return-volatility state	3	33	14.16
S&P 500	0	Adverse return-volatility state	4	24	10.30
S&P 500	0	Adverse return-volatility state	5	23	9.87
S&P 500	0	Adverse return-volatility state	6	12	5.15
S&P 500	0	Adverse return-volatility state	7	13	5.58
S&P 500	0	Adverse return-volatility state	8	6	2.58
S&P 500	0	Adverse return-volatility state	9	3	1.29
S&P 500	0	Adverse return-volatility state	10	1	0.43
S&P 500	0	Adverse return-volatility state	12	1	0.43
S&P 500	1	Intermediate market state	1	147	29.64
S&P 500	1	Intermediate market state	2	120	24.19

Table A12. *Cont.*

Market	Regime	Regime Label	Spell Length (Days)	Frequency	Percentage (%)
S&P 500	1	Intermediate market state	3	102	20.56
S&P 500	1	Intermediate market state	4	65	13.10
S&P 500	1	Intermediate market state	5	23	4.64
S&P 500	1	Intermediate market state	6	14	2.82
S&P 500	1	Intermediate market state	7	8	1.61
S&P 500	1	Intermediate market state	8	6	1.21
S&P 500	1	Intermediate market state	9	3	0.60
S&P 500	1	Intermediate market state	10	2	0.40
S&P 500	1	Intermediate market state	11	2	0.40
S&P 500	1	Intermediate market state	12	3	0.60
S&P 500	1	Intermediate market state	15	1	0.20
S&P 500	2	Favorable return-volatility state	1	37	11.04
S&P 500	2	Favorable return-volatility state	2	43	12.84
S&P 500	2	Favorable return-volatility state	3	31	9.25
S&P 500	2	Favorable return-volatility state	4	39	11.64
S&P 500	2	Favorable return-volatility state	5	36	10.75
S&P 500	2	Favorable return-volatility state	6	32	9.55
S&P 500	2	Favorable return-volatility state	7	32	9.55
S&P 500	2	Favorable return-volatility state	8	21	6.27
S&P 500	2	Favorable return-volatility state	9	19	5.67
S&P 500	2	Favorable return-volatility state	10	8	2.39
S&P 500	2	Favorable return-volatility state	11	3	0.90
S&P 500	2	Favorable return-volatility state	12	8	2.39
S&P 500	2	Favorable return-volatility state	13	8	2.39
S&P 500	2	Favorable return-volatility state	14	6	1.79
S&P 500	2	Favorable return-volatility state	15	4	1.19
S&P 500	2	Favorable return-volatility state	16	3	0.90
S&P 500	2	Favorable return-volatility state	17	1	0.30
S&P 500	2	Favorable return-volatility state	19	2	0.60
S&P 500	2	Favorable return-volatility state	20	1	0.30
S&P 500	2	Favorable return-volatility state	28	1	0.30

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